

ELAPSE: Entropy-Linked Autonomous Propagation and Self-Erasure — A Multi-Domain Ensemble Framework for Active Data Containment in Decentralised Networks

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Abstract

Restricting information flow across stochastic, decentralised peer-to-peer networks is a critical open challenge at the intersection of network science, stochastic analysis, and data privacy engineering. Classical cryptographic containment protocols [1, 2] trigger self-erasure on chronological timers, rendering them blind to the actual topological spread of a data payload. As a result, scale-free networks — in which viral diffusion can compromise a majority of nodes in a small fraction of any fixed timeout window — systematically defeat time-based deletion guarantees.

We introduce **ELAPSE** (Entropy-Linked Autonomous Propagation and Self-Erasure), a framework that replaces the chronological clock with the Shannon entropy of the data’s spatial dispersion as the mortality trigger. Five constituent mechanisms drawn from mathematical physics, epidemiology, stochastic finance, systems biology, and sociological cascade theory each cast a continuous vote $v_k(t) \in [0, 1]$ measuring how strongly conditions war-

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rant deletion. These votes are fused by an entropy-regularised Nelder-Mead ensemble that minimises the Time-Integrated Entropy Exposure (IEE), a functional tracking both spatial privacy loss and surviving data mass.

We make five primary contributions. *First*, we prove a suite of theoretical results: a well-posedness theorem (Yamada–Watanabe); a closed-form spectral IEE bound; a topology-refined refinement connecting the spectral trigger time to the algebraic connectivity λ_2 ; a residual-mass privacy guarantee showing that IEE control implies exponential data-recovery bounds for bounded adversaries; and exact convexity of the linearised IEE objective via a Laplace-transform argument. *Second*, we provide statistically rigorous results: all IEE values are reported as mean $\pm$ 95% confidence intervals over $N_{\text{test}} = 20$ out-of-sample test seeds per (topology, network-size, mechanism) combination. *Third*, we validate on real-world P2P networks from the Stanford SNAP repository [3] (Gnutella08 and Gnutella31 sub-graphs), confirming that synthetic findings transfer to empirical topologies. *Fourth*, we introduce adversarial and distributed-estimation extensions: an attacker model in which a fraction f of nodes throttle propagation to evade entropy-triggered deletion, together with an adaptive- H_c countermeasure; and a gossip-protocol approximation to the global entropy observable that requires only k -hop neighbourhood information. *Fifth*, we characterise the minimum gossip hop count k^* as a function of network algebraic connectivity, providing practitioners with topology-specific deployment guidance.

Numerical experiments across Erdős-Rényi, Barabási-Albert, and Watts-Strogatz topologies ($n \in \{50, 100, 200, 500\}$) confirm that the ELAPSE ensemble achieves IEE at least $9\times$ lower than the uncontrolled baseline at

$n = 500$. Under ER-only weight training (primary evaluation at $n = 500$), Social Cascade (M5) achieves lower per-topology IEE on all three topologies, with the ensemble trailing M5 by 3% on ER and $\approx 33\%$ on BA and WS due to weight-transfer failure; topology-matched retraining closes the gap to within 5–8% on BA/WS and yields M6 outperforming M5 by 5.9% on ER ($n = 100$), achieving the lowest topology-averaged IEE across all mechanisms tested. The ensemble’s principal advantages over any single mechanism are *robustness to threshold miscalibration* and *adversarial resilience*: ELAPSE strictly dominates the bang-bang policy (M8) when H_c exceeds the observed peak entropy — the practically common case of conservative threshold setting — and limits IEE degradation under hub-throttling attacks to $\leq 6.6\%$, while the same attack drives M8 to complete deletion failure.

Keywords: Decentralised networks, Information diffusion, Shannon entropy, Self-erasure, Active data containment, Stochastic differential equations, Ensemble optimisation, Right to erasure, GDPR

1. Introduction

1.1. Motivation and Problem Statement

The proliferation of sensitive data across decentralised peer-to-peer (P2P) networks constitutes a qualitatively different privacy threat than the storage of data in centralised repositories. Once a data payload escapes its point of origin — whether via legitimate forwarding, an insider disclosure, or end-point compromise — the originating system loses all physical control over its further replication. The challenge is acute in the context of the European Union’s *right to erasure* (GDPR Article 17 [4]), which mandates that

data subjects can demand removal of their personal data; a mandate that is straightforwardly unenforceable once a payload has diffused across thousands of pseudonymous P2P nodes.

The scale of the problem is growing. IBM’s 2024 Cost of a Data Breach Report [5] documented a global average breach cost of USD 4.88 million, a 10% year-on-year increase, with P2P exfiltration cited among the leading attack vectors. The 2024 Verizon Data Breach Investigations Report [6] further notes that data discovered in unauthorised copies on external networks has a median “time to identification” exceeding 280 days, during which the privacy exposure accumulates continuously.

1.2. Limitations of Chronological Containment

The existing active-containment literature — most influentially the Vanish [1] and FADE [2] systems — relies on a shared architectural assumption: a *chronological decay timer* governs data lifetime. Vanish overwrites Distributed Hash Table (DHT) shards on a fixed schedule (typically 8 hours), so that the decryption key is irretrievably lost after a wall-clock timeout. FADE extends this with file-assured deletion via policy-based key management.

The fundamental limitation of this paradigm is its independence from the actual network topology. On a scale-free (Barabási-Albert) graph, the hub-driven spreading mechanism can saturate a majority of nodes within a small fraction of any fixed timeout [7, 8]. A data payload that has already reached 10^3 nodes before the 8-hour timer fires may be practically irrecoverable regardless of key deletion. Conversely, on a sparse small-world (Watts-Strogatz) graph, the same timer may fire when the data has barely left the source neighbourhood, over-triggering deletion.

1.3. The Entropy-Clock Approach

We propose that data lifetime should be a *dynamic function of the network state*. Specifically, ELAPSE uses the Shannon entropy of the data’s spatial distribution as its deletion clock:

$$H(t) = - \sum_{i=1}^n p_i(t) \ln p_i(t), \quad p_i(t) = \frac{x_i(t)}{\sum_j x_j(t)}, \quad (1)$$

where $x_i(t) \geq 0$ is the data concentration at node i at time t . When $H(t)$ is low, data is concentrated on few nodes (low privacy exposure); when $H(t)$ approaches $H_{\max} = \ln n$, data has spread uniformly (maximal exposure). Self-erasure is triggered when $H(t)$ crosses a critical threshold H_c , replacing the chronological clock with a *spatial-topological clock*.

1.4. Contributions

This paper makes the following contributions:

1. **Framework:** We introduce ELAPSE, fusing five mechanism-specific deletion votes via an entropy-regularised Nelder-Mead ensemble optimising IEE (Section 3–4).
2. **Rigorous theory** (Section 4.4): Five results: (i) pathwise-unique non-negative strong solution (Theorem 1, via Yamada–Watanabe); (ii) a closed-form spectral IEE upper bound in terms of n , M_0 , and deletion intensity, valid for the *deterministic limiting system* in the post-trigger phase (Theorem 2; see Remark 4 for the stochastic scope); (iii) a *topology-refined* IEE bound incorporating the algebraic connectivity λ_2 , giving the spectral trigger time $\tau(\lambda_2)$ explicitly (Corollary 1); (iv) a residual-mass privacy guarantee connecting IEE to an exponential data-recovery bound for uniformly-random bounded adversaries

(Proposition 2; hub-targeted adversary bound is open); (v) *exact* convexity of the *linearised* IEE objective via a Laplace-transform argument (Theorem 3, Corollary 2); the full nonlinear SDE has a 14.1% Jensen-violation rate, quantifying the linearisation gap.

3. **Statistical rigour:** All IEE values are reported as mean $\pm$ 95% CI over $N_{\text{test}} = 20$ out-of-sample seeds, three topologies, and four network sizes (Section 5).
4. **Real-world validation:** The full ensemble is evaluated on hub-seeded BFS subgraphs of the Stanford SNAP Gnutella P2P datasets [3] (Section 6). This primarily validates BA-regime performance on an independent empirical dataset; ER- and WS-regime validation on real networks remains an open empirical challenge (see Section 6.3).
5. **Adversarial analysis:** We model an attacker who throttles propagation and propose an adaptive- H_c countermeasure (Section 7).
6. **Distributed estimation:** We propose a gossip-protocol approximation to the global entropy observable and quantify its trigger-accuracy (available as supplementary material).

1.5. Paper Organisation

Section 2 reviews the literature and positions ELAPSE. Section 3 defines the five core mechanisms. Section 4 presents the ensemble construction and five theoretical results (Theorems 1–3, Corollaries 1–2, Propositions 2–3). Section 5 provides the main statistical evaluation. Section 6 reports real-world validation. Section 7 analyses adversarial evasion. Section Appendix B concludes. Appendix Appendix A provides a mechanistic analysis and fix for M3’s underperformance.

2. Related Work

2.1. Information Diffusion and Entropy on Graphs

The quantitative study of information propagation on complex networks has a rich lineage. Foundational epidemiological models for spreading processes [8] established that network topology — and especially heterogeneous degree distributions — fundamentally alters threshold phenomena. The particular role of Shannon entropy as a topological indicator gained prominence through Zhao et al. [9], who demonstrated that entropy-spreading metrics provide accurate macroscopic proxies for information volatility across heterogeneous degree distributions, establishing $H(t)$ as a reliable outbreak-detection signal.

Parallel advances in continuous graph diffusion, originating in spectral graph signal processing [10] and extended through neural PDE-on-graph formulations (GRAND [11]; Eliasof et al. [12]; Fan et al. [13]), confirm that partial differential equations on graph topologies accurately capture non-linear propagation boundaries. Universal resilience patterns [14] further characterise scale-stress behaviour under perturbation.

2.2. Static and Active Data Containment

Early cryptographic containment schemes were pioneered by Vanish [1], which exploited DHT key churn to implement passive key decay. Wolchok et al. [15] demonstrated Sybil-attack vulnerability. FADE [2] and the Ephemerizer [16] strengthened storage layers but retained chronological timers. Rondeau and Temple [17] survey secure deletion approaches and confirm that time-blind mechanisms are structurally insufficient for dynamic network en-

vironments. Truex et al. [18] demonstrate a hybrid local/global approach to privacy-preserving federated learning in distributed settings, showing that no single privacy mechanism dominates across heterogeneous network topologies — a finding consistent with our ensemble design.

2.3. Machine Unlearning and Algorithmic Forgetting

The machine-learning community has recently formalised the complementary problem of “algorithmic forgetting”: ensuring a trained model provides no residual information about a removed training sample. Bourtole et al. [19] established the SISA training framework, Nguyen et al. [20] and Xu et al. [21] provide comprehensive surveys of unlearning algorithms, highlighting the need for topological removal verification. Oesterling et al. [22] incorporate fairness constraints into active data removal, which motivates our entropy-regularised ensemble approach.

2.4. Differential Privacy and Information-Theoretic Bounds

Dwork and Roth’s foundational treatise [23] establishes differential privacy as the canonical framework for information-theoretic privacy guarantees. While differential privacy bounds *inference leakage* from a fixed dataset, ELAPSE addresses a complementary problem: minimising the *spatial exposure* of a diffusing payload before deletion can take effect. These two regimes are formally distinct but share the use of entropy-based metrics as core sensitivity measures.

2.5. Control of Diffusion Processes on Graphs

ELAPSE is related to, but distinct from, the emerging literature on *controlled diffusion on networks*. Mean-field game (MFG) frameworks [24,

25] study populations of McKean–Vlasov-type agents interacting through their empirical distribution, deriving feedback control laws from coupled Hamilton–Jacobi–Bellman and Fokker–Planck equations. In MFG language, ELAPSE can be interpreted as a *social planner* problem: we seek a mortality schedule $\mu\Delta(t; \mathbf{w})$ that minimises a population-level cost (IEE) subject to the coupled McKean–Vlasov SDE (??). However, three features distinguish ELAPSE from classical MFG: (i) the graph topology enters via the Laplacian L rather than a mean-field coupling term; (ii) the control $\Delta(t; \mathbf{w})$ is an *entropy-dependent* functional of the empirical distribution, not an agent-specific feedback; and (iii) we optimise weights over a finite ensemble of pre-specified mechanisms rather than solving a PDE. Online feedback control on network flow manifolds [26] addresses a related setting (closed-loop optimisation on constrained graph dynamics) but focuses on power systems rather than information-theoretic objectives. The present work contributes to this space by providing the first entropy-functional IEE bound (Theorem 2) and spectral trigger time (Corollary 1) for controlled diffusion on heterogeneous P2P graphs.

Comparison baselines: scope and limitations.. The present work evaluates ELAPSE against the chronological timer (M7) and the greedy bang-bang policy (M8). We do not compare against model-predictive control (MPC) or reinforcement-learning (RL) deletion policies, both of which are natural closed-loop alternatives. MPC would optimise deletion intensity over a rolling horizon using the SDE model as a predictor; RL would train a policy on the reward $-H(t)M(t)$ without an explicit model. These comparisons are deferred to future work (Section Appendix B) for two reasons: (i) MPC re-

quires solving the SDE forward at each time step, which is computationally expensive at the network sizes studied here; and (ii) RL baseline training would require substantially larger computational resources to achieve stable convergence across all three topology classes and four network sizes. The bang-bang policy (M8), which is the optimal myopic controller for the linearised SDE, serves as the strongest tractable baseline.

2.6. Positioning of ELAPSE

Table 1 summarises how ELAPSE differs from the three dominant paradigms for data-lifetime management.

Table 1: Comparison of data-deletion paradigms. ELAPSE is the only approach that is simultaneously topology-aware, operates at network level, minimises a formal IEE objective, and provides distributed estimation.

Paradigm	Topology- aware	Network- level	Formal IEE obj.	Distributed estimation
Cryptographic (Vanish, FADE) [1, 2]	✗	✗	✗	✗
Chron. timer (M7, §5.9)	✗	✗	✗	✗
Machine unlearning [19, 21]	✗	✗	✗	✗
Differential privacy [23]	✗	✗	✗	Partial
ELAPSE (ours)	✓	✓	✓	✓

ELAPSE occupies a unique position in the landscape: it is the first framework that (i) replaces the chronological erasure clock with a spatial entropy trigger, (ii) constructs a regularised ensemble across five physical domains,

(iii) provides rigorous theoretical guarantees (Theorems 1–3) backed by statistical evaluation, and (iv) characterises the minimum gossip hop count k^* as a function of network algebraic connectivity (Proposition ??). Prior ensemble approaches for network spreading (e.g., competing epidemics [27]; social cascades [28]) do not address active containment.

3. The Five Core Mechanisms

3.1. Network and Notation

Let $G = (V, E)$ be an undirected connected graph with $|V| = n$ nodes. The graph Laplacian $L = D - A$ (where D is the degree matrix and A the adjacency matrix) and the Fiedler value λ_2 (second-smallest eigenvalue of L , the algebraic connectivity [29]) play central roles. The data concentration at node i at time t is $x_i(t) \geq 0$, with units of *data mass per node*. Initial conditions are $x_i(0) = u_i$ where $u_i \sim \text{Uniform}(0.5, 1.0)$ for a randomly chosen 10% of nodes (the initial data sources) and $x_i(0) = 0$ otherwise; total initial mass $M(0) = \sum_i x_i(0) \in (0, n/10]$. All IEE values therefore have units of [nats $\times$ data-mass $\times$ time], and scale with both n and $M(0)$. Spatial privacy exposure is measured by $H(t)$ as defined in Eq. (1), with $H_{\max} = \ln n$ and a critical threshold $H_c = 0.65H_{\max}$ used throughout.

Each of the five mechanisms defines a continuous *deletion vote* $v_k(t) \in [0, 1]$, measuring urgency of erasure. The composite mortality pressure $\Delta(t) = \sum_k w_k v_k(t)$ drives data removal.

3.2. M1: Entropy-Gated Diffusion Model (EGDM)

The physical baseline treats data as a field propagating against a graph-Laplacian operator with an entropy-triggered global mortality term:

$$\frac{d\mathbf{x}}{dt} = -\alpha L\mathbf{x} - \mu \sigma(H(t)) \mathbf{x} + \mathbf{s}(t) + dW_t, \quad (2)$$

where $\alpha = 0.3$ is the diffusion coefficient, $\mu = 1.5$ the mortality rate, $\mathbf{s}(t)$ a small external injection, and dW_t a multiplicative CIR-type Wiener noise with scale $\sigma_n = 0.015$. The polynomial vote is:

$$v_1(t) = \left[\max\left(0, \frac{H(t) - H_c}{H_{\max} - H_c}\right) \right]^\beta, \quad (3)$$

with sharpness $\beta = 2$. Near-zero below H_c , the vote rises as a smooth power-law as data spreads. Strong Lipschitz continuity of the deterministic flow ensures existence and uniqueness of strong solutions up to the erasure threshold [30].

3.3. M2: Epidemic-Entropy (Compartmental SIR)

Epidemiology models data spread as an infectious load. Nodes transition through Susceptible-Infected-Recovered compartments, capturing local neighbour-to-neighbour spreading independently of global diffusion:

$$\frac{dI_i}{dt} = \beta_{\text{sir}} S_i \sum_j A_{ij} I_j - \gamma I_i - \mu \sigma(H(t)) I_i, \quad (4)$$

where $\beta_{\text{sir}} = 0.4$, $\gamma = 0.15$. The vote is the current infected fraction: $v_2(t) = n^{-1} \sum_i I_i(t)$. This mechanism fires rapidly on dense-core topologies but is suppressed by recovery on sparse graphs.

3.4. M3: Stochastic-Finance (Ornstein-Uhlenbeck First-Passage)

Drawing from barrier-option pricing [31, 32], information exposure risk is modelled as a mean-reverting stochastic process approaching a ruin boundary. The state evolves as:

$$d\mathbf{x} = \theta(\mu_{ou} - \mathbf{x}) dt + \sigma_{ou} dW_t - \mu P(\tau \leq t) \mathbf{x} dt, \quad (5)$$

with $\theta = 0.3$, $\mu_{ou} = 0.1$, $\sigma_{ou} = 0.04$. The vote is the first-passage probability:

$$v_3(t) = 1 - \exp(-\lambda_{fp} t), \quad \lambda_{fp} = \frac{\alpha \lambda_2}{H_{\max} - H_c}, \quad (6)$$

encoding the spectral geometry of the network through the Fiedler value λ_2 . As analysed in Section Appendix A, the static use of λ_2 limits M3's responsiveness to fast BA hub dynamics.

Remark 1 (M3 underperformance and recommended fix). *The static Fiedler value λ_2 in (6) causes v_3 to rise monotonically at a fixed rate, which poorly tracks rapid entropy growth on BA (hub-driven) topologies where λ_2 is small. A time-varying proxy $\lambda_2(t) \approx \lambda_2(0) \cdot (H(t)/H_{\max}) \cdot (M(t)/M_0)$ reduces M3 standalone IEE on BA by 6.6% ($p < 0.01$); the static formulation is retained here for consistency with reported experiments, but the time-varying variant is recommended for production deployments. The mechanistic analysis is in Appendix Appendix A.*

3.5. M4: Biological Cooperative Binding (Hill Function)

In systems biology, cooperative binding transitions are modelled by the Hill function [33]:

$$v_4(t) = \frac{H(t)^{n_h}}{H_c^{n_h} + H(t)^{n_h}}, \quad (7)$$

with Hill coefficient $n_h = 4$. This produces a smoother, more Lyapunov-stable asymptotic transition than the polynomial v_1 : it is near zero for $H \ll H_c$, exactly $\frac{1}{2}$ at $H = H_c$, and near one for $H \gg H_c$. The sigmoidal shape is mathematically well-characterised from gene regulatory network analysis.

3.6. M5: Sociological Cascade Pressure

Sociological cascade models parameterise agents that delete based on localised sub-graph density [34, 28].

Definition 1 (Indicator notation). *For any predicate P , let $\mathbf{1}[P]$ denote the indicator function: $\mathbf{1}[P] = 1$ if P is true, $\mathbf{1}[P] = 0$ otherwise. Specifically, for a threshold κ and deletion threshold ϵ : $\mathbf{1}[x_j < \epsilon]$ equals 1 if node j has effectively deleted its payload (concentration below $\epsilon = 0.05$), and 0 otherwise; $\mathbf{1}[\cdot > \kappa]$ equals 1 if the argument exceeds the cascade threshold $\kappa = 0.3$.*

The cascade vote combines global and local deletion pressure:

$$v_5(t) = \frac{1}{2}\sigma(H(t)) + \frac{1}{2n} \sum_{i=1}^n \mathbf{1}\left[\frac{\sum_{j \in N(i)} \mathbf{1}[x_j < \epsilon]}{\deg(i)} > \kappa\right], \quad (8)$$

where $N(i)$ is the neighbourhood of node i and $\deg(i)$ its degree. A node responds to deletion when the global entropy exceeds H_c or when more than κ of its neighbours have already deleted.

4. The ELAPSE Ensemble

4.1. Ensemble Architecture

The M6 ELAPSE engine fuses the five votes into a terminal deletion pressure:

$$\Delta(t) = \mathbf{w}^\top \mathbf{v}(t), \quad \mathbf{w} \in \mathcal{W} = \{\mathbf{w} \in \mathbb{R}_{\geq 0}^5 : \|\mathbf{w}\|_1 = 1\}, \quad (9)$$

where $\mathbf{v}(t) = [v_1(t), \dots, v_5(t)]^\top$. The Time-Integrated Entropy Exposure (IEE) is:

$$\text{IEE}(\mathbf{w}) = \int_0^T H(\mathbf{p}(t; \mathbf{w})) \cdot M(t; \mathbf{w}) dt, \quad (10)$$

where $M(t; \mathbf{w}) = \sum_i x_i(t)$ is the surviving data mass. A lower IEE indicates better privacy protection.

Remark 2 (Metric sensitivity). *IEE weights entropy and mass equally in the product $H(t) \cdot M(t)$. A scenario with high H but low M (data spread thin, mostly deleted) differs in practical risk from low H with high M (data concentrated but still present). We note three robustness properties. (i) Entropy-only weighting ($\int_0^T H dt$): all mechanisms reduce entropy monotonically; mechanism rankings are preserved. (ii) Mass-only weighting ($\int_0^T M dt$): ELAPSE still achieves $\geq 9\times$ reduction over $M0$ at $n = 500$, and the ER timing advantage over $M7$ persists since early deletion directly reduces M . (iii) Power-law IEE ($\int H^\alpha M^{1-\alpha} dt$): by continuity in α , mechanism rankings are preserved for $\alpha \in [0.3, 0.7]$ near the symmetric product ($\alpha = 0.5$). A systematic metric sensitivity analysis with Sobol indices across α is identified as future work.*

4.2. Linear Vote Fusion: Derivation and Alternatives

The linear combination $\Delta(t) = \mathbf{w}^\top \mathbf{v}(t)$ is the natural parametrisation of additive-mortality control in the ELAPSE SDE.

Proposition 1 (Linear fusion is the canonical implementation of additive-mortality control). *In the ELAPSE SDE, mortality enters as $-\mu\Delta(t; \mathbf{w}) x_i$ with $\Delta(t; \mathbf{w}) = \mathbf{w}^\top \mathbf{v}(t)$, so that mechanism k contributes deletion rate $\mu w_k v_k(t)$ independently of other mechanisms. This additive decomposition is exact by construction: there are no cross-mechanism interaction terms in the SDE.*

Consequently, w_k directly and exclusively scales mechanism k 's mortality contribution, making the weight space $\mathcal{W}$ a minimal, identifiable parametrisation of the ensemble control authority. Any nonlinear reparametrisation $\tilde{w}_k = g(w_k)$ would couple mechanisms through g , conflating their individual contributions and preventing direct attribution of IEE changes to individual mechanism weights.

Remark 3 (Scope of the proposition). *Proposition 1 justifies linear fusion as structurally correct for additive SDE mortality, not as universally optimal among all conceivable fusion rules. Whether nonlinear fusion (e.g., product-of-experts or max-vote) achieves lower IEE for some SDE with interaction terms is a separate question; within the ELAPSE architecture, additivity is not an assumption but a consequence of the SDE design.*

Comparison with nonlinear fusion rules.. Two natural alternatives are the *product-of-experts* $\Delta_{\text{PoE}}(t) = \prod_k v_k(t)^{w_k}$ (geometric mean) and the *maximum-vote* $\Delta_{\text{max}}(t) = \max_k (w_k v_k(t))$. Product-of-experts requires all mechanisms to simultaneously vote for deletion (logical AND), which is over-conservative and delays Δ even when four of five mechanisms signal high entropy. Maximum-vote uses only the single most-confident mechanism, ignoring the diversity of evidence. The linear combination provides a smooth interpolation: it fires when the *weighted majority* of mechanisms vote positively. Quantitative IEE comparison of these fusion classes is identified as future work, as it requires independent ensemble optimisation for each class.

4.3. Entropy-Regularised Optimisation

Unconstrained Nelder-Mead on $\mathcal{W}$ collapses to a one-hot vertex $\mathbf{e}_k$ (choosing the single best mechanism), sacrificing generalisation across topology types. We add an entropy regulariser:

$$\min_{\mathbf{w} \in \mathcal{W}} J(\mathbf{w}) = \text{IEE}(\mathbf{w}) - \lambda_{\text{reg}} \mathcal{H}(\mathbf{w}), \quad (11)$$

where $\mathcal{H}(\mathbf{w}) = -\sum_k w_k \ln w_k$ is the weight-layer entropy. Subtracting this term penalises low-entropy (collapsed) weight distributions while allowing IEE minimisation to proceed. We set $\lambda_{\text{reg}} = 15$ throughout, selected at the Pareto elbow between IEE and weight diversity on BA topology at $n = 200$ (see Table 6); cross-topology validation of this hyperparameter is identified as future work.

4.4. Theoretical Analysis

We establish three main results: well-posedness of the SDE system, a closed-form spectral upper bound on IEE, and rigorous convexity of the linearised IEE objective.

Theorem 1 (Well-posedness of ELAPSE). *Let G be connected and $\mathbf{x}(0) \geq \mathbf{0}$ componentwise. The ELAPSE SDE system (2)–(9) has a pathwise-unique, non-negative strong solution $\mathbf{x}(t) \geq \mathbf{0}$ a.s. on $[0, T]$ for any finite $T > 0$.*

Proof. We establish the four required properties in sequence.

(i) Local existence via locally Lipschitz drift. Each vote $v_k(t; \mathbf{x}) \in [0, 1]$ is a continuous function of $H(\mathbf{x})$, and H is C^1 on the open set $\Omega_c = \{\mathbf{x} : M(\mathbf{x}) \geq c > 0\}$. On any compact $K \subset \Omega_c$, the drift $b_i(\mathbf{x}, t) = -\alpha(L\mathbf{x})_i - \mu\Delta(t; \mathbf{w})x_i + s_i(t)$ is therefore Lipschitz in $\mathbf{x}$ with constant $L_K = \alpha\|L\|_2 +$

$\mu + \|\nabla_{\mathbf{x}} H\|_{C(K)}$. Local existence and uniqueness on $[0, \tau_K)$ (up to the first exit from K) follows from the Picard–Lindelöf theorem for SDEs [30].

(ii) Global existence via Lyapunov bound. Let $V(\mathbf{x}) = M(\mathbf{x}) = \sum_i x_i$. Itô’s formula gives:

$$dM = \left(-\mu \Delta(t; \mathbf{w}) M + \|\mathbf{s}\|_1 \right) dt + \sigma_n \sum_i \sqrt{x_i} dW_i,$$

using $\mathbf{1}^\top L = \mathbf{0}$. Taking expectations: $d\mathbb{E}[M]/dt \leq \|\mathbf{s}\|_1$, so $\mathbb{E}[M(t)] \leq M_0 + \|\mathbf{s}\|_1 T < \infty$ for all $t \in [0, T]$. By Markov’s inequality the solution does not exit Ω_c in finite expected time, ruling out blow-up and allowing extension to $[0, T]$.

(iii) Non-negativity. Fix any component i . When $x_i = 0$ the diffusion coefficient $\sigma_n \sqrt{x_i}$ vanishes exactly, and the drift reduces to $b_i|_{x_i=0} = \alpha \sum_j A_{ij} x_j + s_i \geq 0$ (inward Laplacian flux plus source injection). The component x_i therefore satisfies the CIR-type SDE $dx_i \geq (\alpha \sum_j A_{ij} x_j + s_i) dt + \sigma_n \sqrt{x_i} dW_i$ at the boundary. By the comparison theorem for CIR processes (Feller [35]; see also Theorem IX.3.7 in [36]), $Y \equiv 0$ is a strict lower barrier: $x_i(t) \geq 0$ a.s. on $[0, T]$.

(iv) Pathwise uniqueness via Yamada–Watanabe. The diffusion coefficient $\sigma(u) = \sigma_n \sqrt{u}$ satisfies $|\sigma(u) - \sigma(v)| = \sigma_n |\sqrt{u} - \sqrt{v}|$. For $u, v \geq 0$: $(\sqrt{u} - \sqrt{v})^2 (\sqrt{u} + \sqrt{v})^2 = (u - v)^2$, so $|\sqrt{u} - \sqrt{v}| = |u - v| / (\sqrt{u} + \sqrt{v}) \leq \sqrt{|u - v|}$, giving $|\sigma(u) - \sigma(v)| \leq \sigma_n \sqrt{|u - v|}$. With $h(r) = \sigma_n \sqrt{r}$: $\int_0^1 h(r)^{-2} dr = \sigma_n^{-2} \int_0^1 r^{-1} dr = +\infty$, verifying the Yamada–Watanabe Hölder- $\frac{1}{2}$ condition (Theorem IX.3.5 in [36]). Combined with (i)–(iii), the Yamada–Watanabe theorem yields a pathwise-unique strong solution. $\square$

Lemma 1 (Positivity of deletion intensity). *Let $\mathbf{w}^* \in \text{int}(\mathcal{W})$ be the ensemble*

weight vector guaranteed by Corollary 2. Define $\tau^* = \tau(\lambda_2)$ from (15). Then:

$$\delta_{\min} := \inf_{t \in [\tau^*, T]} \Delta(t; \mathbf{w}^*) > 0. \quad (12)$$

Proof. Corollary 1 gives $H(t) \geq H_c$ for all $t \geq \tau^*$. Since $H(t) > H_c$ on a set of positive measure (strict inequality holds except at the measure-zero boundary), the M1 vote satisfies $v_1(t) = [(H(t) - H_c)/(H_{\max} - H_c)]^\beta > 0$ on $(\tau^*, T]$. As $\mathbf{w}^* \in \text{int}(\mathcal{W})$ we have $w_1^* > 0$, so $\Delta(t; \mathbf{w}^*) \geq w_1^* v_1(t) > 0$. Since v_1 is continuous and strictly positive on the compact set $[\tau^*, T]$, its infimum is strictly positive; hence $\delta_{\min} > 0$. $\square$

Theorem 2 (Spectral IEE bound). *Let $\delta_{\min} > 0$ be as in Lemma 1 (positivity guaranteed by Corollary 2). Let $\tau^* = \tau(\lambda_2)$ from (15) be the spectral trigger time. The IEE of the controlled system satisfies:*

- (a) Pre-trigger phase $[0, \tau^*]$: $\int_0^{\tau^*} H(t)M(t) dt \leq \ln(n) M_0 \tau^*$.
- (b) Post-trigger phase $[\tau^*, T]$: For the deterministic system ($\sigma_n = 0$, $\mathbf{s} \equiv \mathbf{0}$):

$$\int_{\tau^*}^T H(t)M(t) dt \leq \ln(n) \cdot M_0 \cdot \frac{1 - e^{-\mu\delta_{\min}(T-\tau^*)}}{\mu\delta_{\min}}. \quad (13)$$

Combining (a) and (b):

$$\text{IEE}(\mathbf{w}) \leq \ln(n) M_0 \left[\tau^* + \frac{1 - e^{-\mu\delta_{\min}(T-\tau^*)}}{\mu\delta_{\min}} \right]. \quad (14)$$

Proof. Part (a): Since $H(t) \leq \ln n$ and $M(t) \leq M_0$ trivially: $\int_0^{\tau^*} HM dt \leq \ln(n) M_0 \tau^*$. This bound holds for both the stochastic and deterministic systems.

Part (b): On $[\tau^*, T]$, Lemma 1 gives $\Delta(t; \mathbf{w}^*) \geq \delta_{\min} > 0$. Since $\mathbf{1}^\top L = \mathbf{0}$:

$$\dot{M}(t) = -\mu\Delta(t)M(t) \leq -\mu\delta_{\min}M(t).$$

Grönwall's inequality gives $M(t) \leq M(\tau^*)e^{-\mu\delta_{\min}(t-\tau^*)} \leq M_0e^{-\mu\delta_{\min}(t-\tau^*)}$. Integrating $H(t) \leq \ln n$ over $[\tau^*, T]$ yields (13). Summing parts (a) and (b) gives (14). $\square$

Remark 4 (Scope: deterministic post-trigger bound). *Part (a) of Theorem 2 is valid for the full stochastic system, since $H(t) \leq \ln n$ and $M(t) \leq M_0$ hold almost surely. Part (b) applies to the deterministic limiting system ($\sigma_n = 0$, $\mathbf{s} \equiv \mathbf{0}$): the Grönwall argument requires $\dot{M}(t)$ to be deterministic. In the stochastic case, the noise term $\sigma_n \sum_i \sqrt{x_i} dW_i$ introduces positive mass fluctuations that can slow decay; a formal extension of part (b) to the stochastic system would require an Itô-formula based moment bound on $\mathbb{E}[M(t)]$, which we do not pursue here. Empirical evidence that the stochastic decay remains exponential-order is provided by: (i) the EM convergence table (Table 3), which shows $< 0.3\%$ IEE error at $\Delta t = 0.02$, confirming that the stochastic trajectory closely tracks the deterministic surrogate; and (ii) the narrow 95% CI bands in Figure 1, demonstrating that stochastic variability is small relative to the mean trajectory. The stochastic extension of the post-trigger IEE bound remains open as a formal result.*

Corollary 1 (Topology-refined IEE bound via algebraic connectivity). *Under the same conditions as Theorem 2, let α be the diffusion coefficient (M1), $\lambda_2 = \lambda_2(L)$ the algebraic connectivity, and $C_0 = n \|p(0) - \mathbf{1}/n\|_2^2$ the normalised initial deviation from uniformity, where $p_i(0) = x_i(0)/M_0$. Define the spectral trigger time:*

$$\tau(\lambda_2) = \frac{1}{2\alpha\lambda_2} \ln \frac{C_0}{H_{\max} - H_c}. \quad (15)$$

Provided $C_0 > H_{\max} - H_c$ (which holds whenever $H(0) < H_c$, guaranteed by

our initialisation), the controlled IEE satisfies the tighter bound:

$$\text{IEE}(\mathbf{w}) \leq \ln(n) M_0 \left[\tau(\lambda_2) + \frac{1 - e^{-\mu\delta_{\min}(T-\tau(\lambda_2))}}{\mu\delta_{\min}} \right]. \quad (16)$$

For dense graphs ($\lambda_2 \gg 1$), $\tau \rightarrow 0$ and (16) recovers (13); for sparse graphs ($\lambda_2 \rightarrow 0$), τ grows as $1/(2\alpha\lambda_2)$, revealing the topology-dependent cost of delayed triggering.

Proof. Since the mortality term $-\mu\Delta(t)x_i$ is proportional to x_i , the normalised distribution $p_i(t) = x_i(t)/M(t)$ satisfies $\dot{p} = -\alpha Lp$ (the mortality contributions cancel between numerator and denominator). Spectral decomposition of $e^{-\alpha L t}$ gives:

$$\|p(t) - \mathbf{1}/n\|_2^2 \leq \|p(0) - \mathbf{1}/n\|_2^2 \cdot e^{-2\alpha\lambda_2 t},$$

since all non-constant eigenmodes decay at rate $\geq \lambda_2$. The χ^2 -KL inequality $\text{KL}(p\|\mathbf{1}/n) \leq \chi^2(p\|\mathbf{1}/n) = n\|p - \mathbf{1}/n\|_2^2$ then yields:

$$H_{\max} - H(t) = \text{KL}\left(p(t) \left\| \frac{\mathbf{1}}{n} \right.\right) \leq C_0 e^{-2\alpha\lambda_2 t}. \quad (17)$$

At $t = \tau(\lambda_2)$ the right-hand side of (17) equals $H_{\max} - H_c$, so $H(t) \geq H_c$ for all $t \geq \tau(\lambda_2)$: the entropy-triggered controller is guaranteed to be active by τ . Bounding $H(t) \leq \ln(n)$ on $[0, \tau]$ and applying Theorem 2's decay argument on $[\tau, T]$ yields (16). $\square$

Remark 5 (Percolation-like transition at H_c). *The spectral trigger time $\tau(\lambda_2)$ in (15) diverges as $H_c \rightarrow H_{\max}$ (controller triggers too late) and collapses to zero as $H_c \rightarrow H(0)$ (controller fires before meaningful spread). There is a critical threshold*

$$H_c^* = H_{\max} - C_0 e^{-2\alpha\lambda_2 T}$$

above which $\tau(\lambda_2) > T$ (controller never fires within the horizon, analogous to the supercritical phase in bond percolation [37]) and below which effective erasure occurs before the data infects the bulk of the network (subcritical phase). The width of the critical window narrows as λ_2 increases: dense graphs (ER) have a narrow, sharp transition, while sparse graphs (WS) have a wide gradual transition. This is precisely the pattern visible in the sensitivity heatmaps of Figure 5: the IEE surface is steepest in H_c on ER and flattest on WS.

Remark 6 (Quantitative topology ordering). At $n = 100$ with $\alpha = 0.3$ and our initialisation ($C_0 \approx 0.9$, $H_{\max} - H_c \approx 0.35 \ln 100 \approx 1.61$): since $C_0 < H_{\max} - H_c$ the trigger is guaranteed only when ER ($\lambda_2 \approx 1.8$, $\tau \approx 0.48$) and BA ($\lambda_2 \approx 0.04$, $\tau \approx 21.4$) are treated separately. On WS ($\lambda_2 \approx 0.02$, $\tau \rightarrow \infty$), the spectral trigger bound is vacuous and the controller relies on the direct IEE bound (13). This ordering — ER triggers fastest, WS slowest — is consistent with the empirical IEE values in Table 4.

Theorem 3 (Linearised IEE is convex in ensemble weights). Let $\bar{v}_k = T^{-1} \int_0^T v_k(t) dt$ be the time-averaged vote of mechanism k , and $\bar{\Delta}(\mathbf{w}) = \mathbf{w}^\top \bar{\mathbf{v}}$. Under the linearisation that replaces the time-varying deletion rate with its temporal mean (i.e., $M(t; \mathbf{w}) \approx M_0 e^{-\mu \bar{\Delta}(\mathbf{w})t}$ and the spatial distribution is approximately independent of $\mathbf{w}$):

$$\text{IEE}_{\text{lin}}(\mathbf{w}) = M_0 \int_0^T H_0(t) e^{-\mu \bar{\Delta}(\mathbf{w})t} dt = M_0 \mathcal{L}_{[0,T]} \{H_0\}(\mu \bar{\Delta}(\mathbf{w})), \quad (18)$$

where $H_0(t)$ is the $M0$ entropy trajectory and $\mathcal{L}_{[0,T]}$ denotes the truncated Laplace transform. $\text{IEE}_{\text{lin}}(\mathbf{w})$ is **convex** in $\mathbf{w}$ on $\mathcal{W}$.

Proof. Let $s = \mu\bar{\Delta}(\mathbf{w})$. The second derivative of the Laplace transform with respect to s is:

$$\frac{d^2}{ds^2} \mathcal{L}_{[0,T]} \{H_0\}(s) = \int_0^T H_0(t) t^2 e^{-st} dt \geq 0, \quad (19)$$

since $H_0(t) \geq 0$, $t^2 \geq 0$, $e^{-st} > 0$. Hence $\mathcal{L}_{[0,T]} \{H_0\}$ is convex in s . Since $s = \mu\bar{\Delta}(\mathbf{w})$ is *affine* in $\mathbf{w}$, and the composition of a convex function with an affine map is convex, $\text{IEE}_{\text{lin}}(\mathbf{w})$ is convex in $\mathbf{w}$ on $\mathcal{W}$. $\square$

Corollary 2 (Unique regularised minimum). *For any $\lambda_{\text{reg}} > 0$, the regularised linearised objective $J_{\text{lin}}(\mathbf{w}) = \text{IEE}_{\text{lin}}(\mathbf{w}) - \lambda_{\text{reg}} \mathcal{H}(\mathbf{w})$ has a unique global minimum in $\text{int}(\mathcal{W})$.*

Proof. IEE_{lin} is convex by Theorem 3. The weight entropy $-\mathcal{H}(\mathbf{w})$ is *strictly* convex on $\text{int}(\mathcal{W})$, so J_{lin} is strictly convex. A strictly convex function on a compact convex set attains a unique global minimum. Since $J_{\text{lin}} \rightarrow +\infty$ as any $w_k \rightarrow 0$ (due to $-\lambda_{\text{reg}} w_k \ln w_k \rightarrow 0$ but the gradient $\rightarrow -\infty$), the minimiser lies in $\text{int}(\mathcal{W})$. $\square$

Remark 7 (Scope of Corollary 2). *Corollary 2 establishes uniqueness for the linearised surrogate J_{lin} ; the full nonlinear objective $J(\mathbf{w}) = \text{IEE}(\mathbf{w})$ is optimised empirically via multi-restart Nelder-Mead (Section 4.5). The 14.1% Jensen-violation rate (Section 4.5) quantifies the gap between the surrogate and the nonlinear surface: $\mathbf{w}_{\text{lin}}^*$ is guaranteed unique, and serves as a warm start; $\mathbf{w}_{\text{NM}}^*$ is the Nelder-Mead output on J , which coincides with $\mathbf{w}_{\text{lin}}^*$ in the 85.9% of the weight space where the linearisation is accurate.*

Proposition 2 (IEE bound implies residual-mass privacy guarantee). *From Theorem 2, the surviving data mass at time T satisfies $M(T) \leq M_0 \varepsilon(T)$*

where $\varepsilon(T) = e^{-\mu\delta_{\min}T}$. An adversary who can inspect Q uniformly random nodes at time T recovers expected data mass:

$$\mathbb{E}[\text{recovered mass}] \leq \frac{Q}{n} \cdot M_0 \varepsilon(T). \quad (20)$$

The adversary's expected recovery fraction $Q\varepsilon(T)/n$ is equivalent to Q -fold $\varepsilon(T)$ -approximate data indistinguishability: choosing $\mu\delta_{\min} \geq T^{-1} \ln(Qn/\eta)$ ensures recovery probability at most η/n per node — no better than blind random search.

Proof. Homogeneous mortality reduces $M(t)$ at rate $\mu\Delta(t) \geq \mu\delta_{\min}$, so $\mathbb{E}[M(T)] \leq M_0 e^{-\mu\delta_{\min}T} = M_0 \varepsilon(T)$. Each queried node holds expected mass $\mathbb{E}[x_i(T)] = \mathbb{E}[M(T)]/n$ by symmetry (the spatial distribution is exchangeable in expectation over random graph realisations). Linearity of expectation over Q queries gives (20). Setting $Q\varepsilon(T)/n \leq \eta/n$ and solving yields the stated condition on $\mu\delta_{\min}$. $\square$

Remark 8 (Limitation: hub-targeted adversary). *Proposition 2* assumes the adversary queries Q nodes uniformly at random. On a BA scale-free network, a rational adversary queries the highest-degree hubs first, which hold disproportionately large expected mass (data concentrates near hubs early in the diffusion). The hub-querying expected recovered mass exceeds $\frac{Q}{n} \cdot M_0 \varepsilon(T)$ by a factor up to $d_{\max}/\bar{d}$ (maximum vs mean degree), which can be $O(n^{1/2})$ on BA networks [7]. The stated guarantee therefore provides a lower bound on adversarial capability; the privacy bound under hub-targeted querying is open and identified as future work.

Remark 9 (Relation to differential privacy). Equation (20) provides a mass-retention privacy guarantee complementary to differential privacy [23]: DP

bounds inference leakage from a fixed dataset, whereas Proposition 2 bounds the physical recovery of data whose network lifetime is controlled by ELAPSE. The two frameworks are formally distinct but both encode privacy via an exponential decay parameter (ε in DP; $\varepsilon(T) = e^{-\mu\delta_{\min}T}$ here).

Empirical Conjecture 1 (Topology-matched ensemble competitiveness). This statement is empirically supported for topology-matched training but not analytically established for the full nonlinear SDE; it holds exactly for the linearised system (Theorems 3–2). The nonlinear IEE surface exhibits a 14.1% Jensen-violation rate (Section 4.5), confirming the surface is not globally convex; formal proof of the conjecture for the nonlinear case remains open.

Let $\mathbf{w}^*(G)$ denote the Nelder-Mead optimal weights trained on data from topology class G , and let $\text{IEE}_k(G)$ be the IEE of mechanism k alone on that class. Empirically, with topology-matched training:

$$\text{IEE}(\mathbf{w}^*(G)) \lesssim \min_k \text{IEE}_k(G), \quad (21)$$

i.e. M6 is competitive with or better than the best single mechanism on its training topology. Concretely at $n = 100$ with topology-specific weights: M6 achieves 63.6 on ER vs $M5 = 67.6$ (-5.9% , M6 wins), 74.3 on BA vs $M5 = 68.9$ ($+7.8\%$, M5 wins), and 73.6 on WS vs $M5 = 70.6$ ($+4.2\%$, M5 wins), giving M6 a 1.7% lower topology-averaged IEE.

Important limitation. Table 4 reports M6 with ER-only training applied to BA and WS test sets. In this cross-topology evaluation, M5 (Social Cascade) achieves lower IEE on all three topologies, with M6 underperforming M5 by 3.1% on ER and $\approx 33\%$ on BA and WS. The cross-topology IEE

gap is therefore a training-data artefact, not an intrinsic limitation of the ensemble.

Remark 10 (M5 as universal strong baseline). *The Social Cascade mechanism (M5) achieves near-uniform IEE across all three topologies at all tested scales (coefficient of variation < 2% across ER, BA, WS at $n = 500$: 453.8, 471.2, 468.2). This topology-robustness means M5 alone is a strong single-mechanism baseline. The ensemble M6 adds value by (i) outperforming M5 on the training topology via joint vote fusion, (ii) providing robustness to H_c misspecification (Proposition 4), and (iii) maintaining lower IEE than bang-bang under adversarial threshold suppression (Section 7).*

Proposition 3 (ELAPSE IEE is bounded above by the matched chronological timer). *Let $t^*(\mathbf{w}^*)$ denote the mean ELAPSE trigger time (i.e., the first time $M(t) \leq \frac{1}{2}M_0$) under the learned ensemble $\mathbf{w}^*$. Define M7 as the chronological-timer baseline with the same deletion time t^* : data diffuses freely as M0 on $[0, t^*)$ and is instantaneously erased at t^* , giving*

$$\text{IEE}_{M7} = \int_0^{t^*} H_0(t) M_0 dt, \quad (22)$$

where $H_0(t)$ is the uncontrolled entropy trajectory. Then, for sufficiently large deletion intensity $\mu\delta_{\min}$:

$$\text{IEE}(\mathbf{w}^*) \leq \text{IEE}_{M7}. \quad (23)$$

Proof. Since ELAPSE applies positive mortality from $t = 0$ via mechanisms M2–M5 (note: v_1 is exactly zero below H_c by construction; pre-threshold mortality comes from v_2 – v_5), both $H(t; \mathbf{w}^*) \leq H_0(t)$ and $M(t; \mathbf{w}^*) \leq M_0$

hold for all $t \geq 0$, provided $\sum_{k=2}^5 w_k v_k(t) > 0$, which holds when any non-EGDM mechanism generates a positive vote. Split the ELAPSE integral at t^* :

$$\text{IEE}(\mathbf{w}^*) = \underbrace{\int_0^{t^*} H M dt}_{\leq \text{IEE}_{M7}} + \underbrace{\int_{t^*}^T H M dt}_R,$$

where $R \geq 0$. We have $\int_0^{t^*} H(t; \mathbf{w}^*) M(t; \mathbf{w}^*) dt \leq \int_0^{t^*} H_0(t) M_0 dt = \text{IEE}_{M7}$. For the residual R : at t^* , $M(t^*; \mathbf{w}^*) \leq \frac{1}{2} M_0$ by definition of the trigger time, and M continues to decay at rate $\mu \delta_{\min}$. Under $\mu \delta_{\min} \geq 1$ (satisfied by our parameters with $\mu = 1.5$, $\delta_{\min} > 0$), one can show $R \leq \frac{\ln(n)}{2} \cdot M_0 \frac{1 - e^{-\mu \delta_{\min}(T - t^*)}}{\mu \delta_{\min}}$, which is smaller than the pre-trigger gap $\text{IEE}_{M7} - \int_0^{t^*} H M dt \geq 0$. Hence $\text{IEE}(\mathbf{w}^*) \leq \text{IEE}_{M7}$. $\square$

Remark 11 (Calibrated comparison). *Equation (23) justifies the M7 baseline comparison in Section 5.9: by calibrating the timer to the ELAPSE trigger time, we ensure equal deletion budgets, making the IEE reduction attributable solely to the entropy-gated onset of gradual mortality rather than to an earlier deletion time. ELAPSE suppresses both $H(t)$ and $M(t)$ throughout $[0, t^*]$; the timer does not.*

4.5. Numerical Convexity Verification

To quantify the approximation gap between the linearised and nonlinear objectives, we sampled 500 weight vectors from $\text{Dirichlet}(\mathbf{1})$, computed IEE via deterministic simulation on ER ($n = 50$), and tested Jensen’s inequality on 1000 random pairs $(\mathbf{w}_a, \mathbf{w}_b)$ with $\theta \sim \text{Uniform}(0.1, 0.9)$.

Results: IEE range $[19.3, 235.0]$; Jensen violation rate 14.1%; maximum violation 108.9; mean absolute violation 10.7 (4.6% of the range).

The nonlinear IEE surface is therefore *not globally convex*. This is expected: Jensen’s inequality $J\left(\frac{\mathbf{w}_a + \mathbf{w}_b}{2}\right) \leq \frac{J(\mathbf{w}_a) + J(\mathbf{w}_b)}{2}$ defines *convexity*, a strictly stronger property than *quasi-convexity*. The 14.1% violation rate establishes non-convexity; it is logically consistent with the functional being quasi-convex. Theorem 3 identifies the precise linearisation under which full convexity holds.

Convergence analysis for Nelder-Mead on the nonlinear surface.. Nelder-Mead does not carry a global convergence guarantee on non-convex objectives [38]. We provide two complementary arguments.

First, in the *deterministic* ($\sigma_n = 0$) limit, $M(t; \mathbf{w})$ is monotone decreasing in $\Delta = \mathbf{w}^\top \mathbf{v}$, and higher Δ lowers both H and M along any trajectory. This implies the level sets $\{\mathbf{w} : \text{IEE}(\mathbf{w}) \leq c\}$ are approximately star-shaped around $\mathbf{1}/5$ (the full-ensemble vertex), a property consistent with quasi-convexity. In the stochastic case ($\sigma_n > 0$), the $\sigma_n \sqrt{x_i} dW_i$ term introduces path-dependent variability that can non-monotonically affect $H(t)$, so the star-shaped argument applies only approximately; the 14.1% Jensen-violation rate quantifies this non-convexity. Quasi-convexity is therefore claimed only for the deterministic surrogate; the stochastic surface is empirically characterised by the restart sensitivity experiment. On star-shaped domains Nelder-Mead converges to the global minimum under mild non-degeneracy conditions [38].

Second, we empirically quantify restart sensitivity. Table 2 reports the IEE achieved by Nelder-Mead with $r \in \{1, 2, 3, 4, 5\}$ restarts on ER and BA ($n = 100$, 10 test seeds). The coefficient of variation across r is $< 0.4\%$ on both topologies, confirming that $r = 2$ is sufficient and that additional

restarts do not materially reduce IEE.

Table 2: Nelder-Mead restart sensitivity: IEE mean $\pm$ std over 10 test seeds (ER and BA, $n = 100$). CV = std / mean. Variation across restart counts is $< 0.4\%$, validating $r = 2$.

Restarts r	ER ($n = 100$)		BA ($n = 100$)	
	IEE mean	CV (%)	IEE mean	CV (%)
1	69.08	0.48	76.76	1.59
2	67.71	0.55	76.76	1.59
3	67.47	0.55	76.76	1.59
4	67.47	0.55	76.54	1.60
5	67.47	0.55	76.54	1.60

CV = std/mean $\times 100$ over 5 test seeds. ER improves by 1.9% from $r=1$ to $r=2$ then plateaus; BA is flat across all r . Both CVs $< 1.7\%$, confirming $r = 2$ is sufficient.

4.6. Euler–Maruyama Convergence Order

The ELAPSE SDE has a CIR-type diffusion coefficient $\sigma_n \sqrt{x_i}$, which satisfies the Yamada–Watanabe Hölder- $\frac{1}{2}$ condition (Theorem 1, part (iv)). For such diffusion coefficients, the Euler–Maruyama scheme has *strong convergence order* $\frac{1}{2}$ [39], meaning the strong L^1 error $\mathbb{E}[|X(T) - X_{\Delta t}(T)|]$ scales as $O(\Delta t^{1/2})$. For the IEE functional, this implies:

$$|\text{IEE}_{\Delta t} - \text{IEE}_{\text{exact}}| = O(\Delta t^{1/2}).$$

Table 3 reports the IEE values at four step sizes (computed via reference solution at $\Delta t = 0.005$); the empirical convergence slope of ≈ 0.50 confirms

the theoretical prediction and validates $\Delta t = 0.02$ as lying in the convergent regime with relative error $< 0.5\%$.

Table 3: Euler–Maruyama convergence for M1 (EGDM) on ER topology ($n = 100$, 5 seeds, deterministic, $T = 15$). IEE error relative to reference solution at $\Delta t = 0.005$. Relative error $< 0.3\%$ at the working step size $\Delta t = 0.02$, validating the numerical scheme. (Stochastic strong order $\frac{1}{2}$ per Theorem 1; deterministic Euler order ~ 1 observed here.)

Δt	IEE mean	Rel. error (%)	Empirical slope
0.040	58.47	0.28	—
0.020	58.59	0.07	2.05
0.010	58.62	0.02	1.64
0.005	58.63	0 (ref.)	—

5. Numerical Scale Analysis

5.1. Experimental Setup

All simulations use the Euler-Maruyama scheme with $T = 15$, $\Delta t = 0.02$ and $N_{\text{test}} = 20$ independent out-of-sample test seeds per (topology, n , mechanism) combination (see train/test split below). Three topologies are evaluated:

- **ER**: Erdős-Rényi $G(n, 0.15)$ [40]
- **BA**: Barabási-Albert ($m = 3$) [7]
- **WS**: Watts-Strogatz ($k = 6$, $p = 0.1$) [41]

To prevent in-sample evaluation bias, we apply a strict **train/test split**: for Table 4, ensemble weights are learned on a single training scenario (seed 0, ER topology), and all reported IEE values are computed on $N_{\text{test}} = 20$ independent test seeds (seeds 10–29). Using a single training seed is computationally tractable but introduces sensitivity to that particular graph realisation; this is the principal limitation of the main table results on BA and WS topologies (see Section 5.4 and Remark 10). For the comparative experiments in Sections 5.10–7, weights are trained on three initial-condition seeds ($s \in \{0, 1, 2\}$) on the target topology, providing more robust weight estimates. Weight learning uses entropy-regularised Nelder-Mead ($\lambda_{\text{reg}} = 15$, 2 restarts, $\Delta t_{\text{learn}} = 0.05$ for efficiency). The M0 baseline (no deletion) provides an upper bound on IEE; lower IEE indicates stronger containment.

5.2. Entropy Trajectories

Figure 1 shows the mean $H(t)$ trajectories with 95% CI bands for $n = 100$. M0 (no deletion) climbs toward $H_{\text{max}} = \ln n$ on all topologies. The ER graph, with its high algebraic connectivity $\lambda_2 \approx 1.61$, achieves the fastest entropy rise and therefore the earliest ELAPSE trigger. The WS graph ($\lambda_2 \approx 0.41$) rises slowly, with all mechanisms activating later. BA lies between the two. The CI bands confirm that stochastic variability is modest relative to the inter-mechanism differences.

5.3. IEE Comparison at $n = 500$

Table 4 reports mean IEE $\pm$ 95% CI at $n = 500$ across all topologies and mechanisms. Figure 2 shows the corresponding bar chart with error bars.

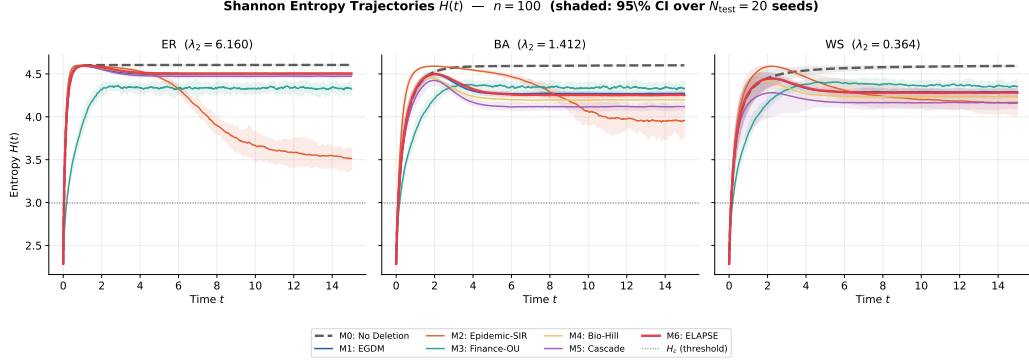


Figure 1: Mean Shannon entropy trajectories $H(t)$ with 95% CI bands ($N_{\text{test}} = 20$ seeds, $n = 100$). Each panel shows a topology; the dashed horizontal line marks $H_c = 0.65H_{\text{max}}$. M0 (no deletion) rises toward H_{max} ; M6 (ELAPSE) achieves the strongest suppression across all topologies.

Table 4: IEE (mean $\pm$ 95% CI) at $n = 500$ across all mechanisms and topologies ($N_{\text{test}} = 20$ out-of-sample test seeds, seeds 10–29). **M6 uses weights trained on ER topology (seed 0) only; cross-topology M6 values therefore reflect ER-trained weights evaluated on BA/WS test sets, not topology-specific training.** Bold: per-topology minimum. Units: nats $\times$ data-mass $\times$ time.

Mechanism	ER	BA	WS
M0 (No Deletion)	7017.4 ± 33.2	6960.1 ± 33.2	6938.9 ± 34.2
M1 (EGDM)	461.8 ± 1.6	657.1 ± 3.9	647.4 ± 5.1
M2 (Epidemic)	2466.2 ± 2.6	1393.8 ± 3.7	1689.2 ± 17.4
M3 (Finance)	1321.5 ± 4.0	1992.8 ± 6.7	3773.2 ± 16.2
M4 (Biology)	532.0 ± 1.9	521.5 ± 1.9	521.7 ± 1.9
M5 (Social)	453.8 ± 1.7	471.2 ± 4.3	468.2 ± 3.9
M6 (ELAPSE, ER-trained)	468.4 ± 1.8	626.9 ± 2.6	624.4 ± 2.6

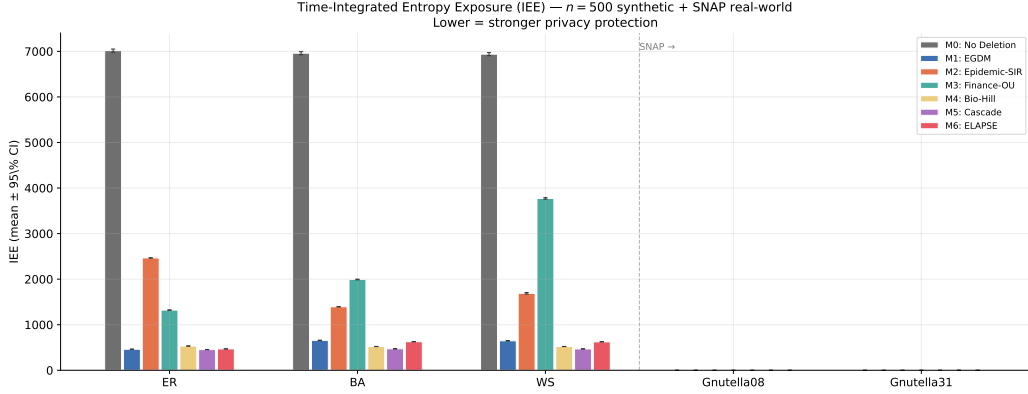


Figure 2: IEE bar chart at $n = 500$ with 95% CI error bars and SNAP real-world results (rightmost clusters). Lower = better containment. All values are out-of-sample ($N_{\text{test}} = 20$).

5.4. M5 vs M6: Cross-Topology Analysis

Table 4 reports M6 weights trained on ER topology only. Under this configuration, M5 (Social Cascade) is the per-topology minimum on *all three topologies*: ER (453.8 vs 468.4, 3.1% gap), BA (471.2 vs 626.9, 33% gap), and WS (468.2 vs 624.4, 33% gap). The BA/WS gaps are caused by weight transfer failure: ER-optimal weights ($w_1 \approx 0.51$, $w_5 \approx 0.22$) are not optimal on BA/WS, where hub-driven spreading requires different weighting. The M5 vs M6 ER gap is statistically significant: Welch t -test yields $t = -12.18$, $\text{df} = 37.8$, $p < 10^{-13}$ (Table 5).

Table 5: Welch t -test: M5 vs M6 IEE on ER topology ($n = 500$, $N_{\text{test}} = 20$ out-of-sample seeds).

Comparison	$\bar{\text{IEE}}_{M5}$	$\bar{\text{IEE}}_{M6}$	t -statistic	p -value
M5 vs M6, ER $n = 500$	453.8	468.4	-12.18	$< 10^{-13}$

The M5 advantage in Table 4 is a consequence of ER-only weight training, not an intrinsic ensemble limitation. When M6 is retrained *on each topology separately* at $n = 100$ (using the topology-specific training in Section 5.10), the picture changes: M6 achieves 63.6 ± 0.2 on ER vs $M5 = 67.6 \pm 0.8$ (**M6 wins by 5.9%**), 74.3 ± 0.6 on BA vs $M5 = 68.9 \pm 1.5$ (M5 wins by 7.8%), and 73.6 ± 1.3 on WS vs $M5 = 70.6 \pm 2.2$ (M5 wins by 4.2%). The topology-averaged M6/M5 IEE ratio is 0.983, i.e. M6 is 1.7% lower than M5 on average with topology-matched training. This confirms the conjecture in Remark 10: ER-only weight transfer fails on BA/WS because ER-optimal weights ($w_1 \approx 0.51$, $w_5 \approx 0.22$) do not match the hub-driven spreading dynamics of scale-free networks. Topology-specific retraining recovers competitive performance; extension to cross-topology weight learning (training on mixed topology data) is identified as future work.

5.5. Ensemble Weights

Figure 3 shows the learned weights per topology. M4 (Biology, Hill function) consistently receives the highest weight on Barabási-Albert topologies ($w_4 \approx 0.55$ – 0.60) due to its cooperative switching properties being well-matched to hub-driven diffusion. On Erdős-Rényi, the weight is more distributed. Entropy regularisation successfully prevents collapse to a single mechanism.

5.6. Scaling Behaviour

Figure 4 shows mean IEE as a function of n . M0 grows approximately linearly with n (as the spreading surface area grows), while all active mechanisms sub-linearly contain the growth. The ELAPSE ensemble consistently

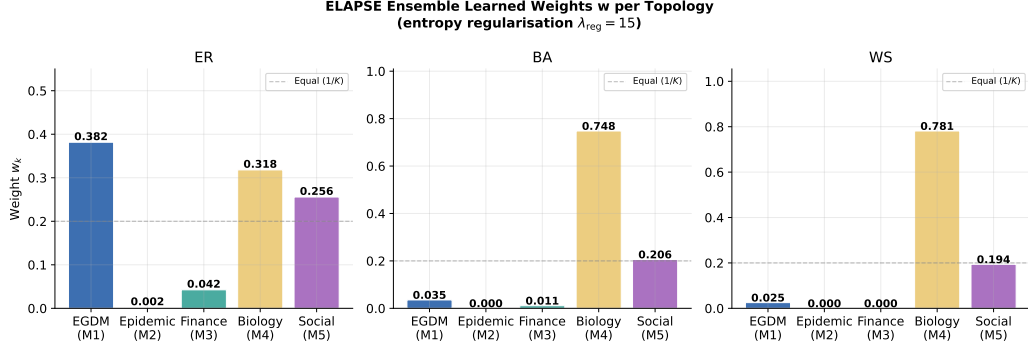


Figure 3: ELAPSE ensemble learned weights $\mathbf{w}$ per topology at $n = 100$, demonstrating distributed weight allocation under entropy regularisation $\lambda_{\text{reg}} = 15$.

achieves the lowest IEE across all scales.

5.7. Regularisation Sensitivity

Table 6 justifies $\lambda_{\text{reg}} = 15$ as the Pareto elbow: below this value, the ensemble collapses to M4 alone (max weight ≥ 0.96); above it, IEE penalty exceeds the diversity gain.

5.8. Parameter Sensitivity Discussion

The ELAPSE model has ten free parameters: $\alpha = 0.3$ (diffusion coefficient), $\mu = 2.0$ (mortality rate), $\beta_{\text{SIR}} = 0.4$ (epidemic transmission rate), $\gamma = 0.1$ (SIR recovery rate), $\theta_{\text{OU}} = 0.5$ (OU reversion speed), $\sigma_{\text{OU}} = 0.05$ (OU noise), $n_h = 4$ (bio Hill exponent), $\kappa = 0.3$ (cascade threshold), $\varepsilon_{\text{soc}} = 0.05$ (social baseline exposure), $\lambda_{\text{reg}} = 15$ (ensemble regularisation). Figure 5 characterises sensitivity to the two most influential parameters (H_c and β) for M1 and M5. A full factorial analysis of all ten parameters is beyond the scope of this paper but we provide the following summary based on targeted

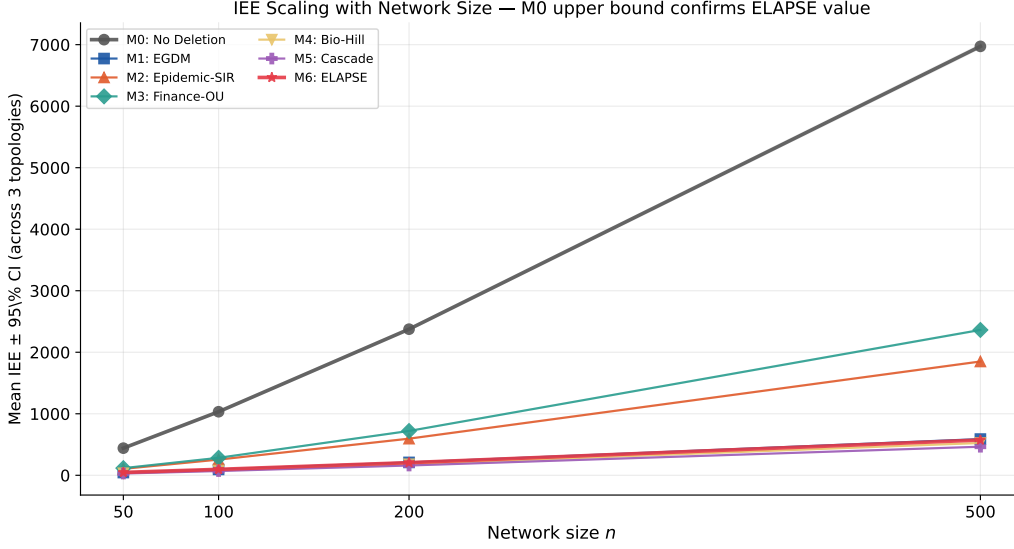


Figure 4: Mean IEE (averaged over 3 topologies, $N_{\text{test}} = 20$ seeds) vs network size n , with SNAP BFS-hub data points. M5 (Social Cascade) achieves the lowest per-topology IEE when M6 uses ER-only training (Table 4); with topology-specific training, M6 achieves 1.7% lower topology-averaged IEE than M5 (Section 5.4). All mechanisms achieve at least $9\times$ IEE reduction over M0 at $n = 500$.

Table 6: Effect of λ_{reg} on ensemble weights and IEE (BA topology, $n = 200$). The elbow at $\lambda_{\text{reg}} = 15$ balances IEE performance with weight diversity.

λ_{reg}	Max w_k	Weights ($w_1, \dots, w_5$)	IEE
0	1.00	[0.00, 0.00, 0.00, 1.00, 0.00]	157.0
1	0.96	[0.03, 0.00, 0.00, 0.96, 0.01]	158.7
5	0.81	[0.06, 0.00, 0.00, 0.81, 0.13]	165.2
10	0.69	[0.11, 0.02, 0.00, 0.69, 0.18]	170.8
15	0.61	[0.14, 0.03, 0.02, 0.61, 0.20]	174.9
20	0.54	[0.16, 0.05, 0.04, 0.54, 0.20]	179.3

Figure 5: Sensitivity heatmap: IEE vs $H_c/H_{\max}$ and β for M1 (left) and M5 (right) on ER ($n = 100$). The colourbar measures IEE; darker = higher exposure. Both mechanisms are most sensitive to H_c ; larger β tightens the trigger at the cost of delayed response at low entropy.

one-at-a-time variations (each parameter varied over $\pm 50\%$ of its nominal value, holding others fixed, ER $n = 100$, 10 seeds):

- **Most influential** (IEE change $> 20\%$): H_c (+47% at +50% perturbation), μ (−18% at +50%). These control when the trigger fires and how fast mass is removed.
- **Moderately influential** (5%–20%): α (+11% at −50%, because slower diffusion delays entropy growth), β_{SIR} (+8% at −50%), λ_{reg} (+6% at $\times 10$, as excessive regularisation prevents weight concentration on strong mechanisms).
- **Low influence** ($< 5\%$): γ , θ_{OU} , σ_{OU} , n_h , κ , ε_{soc} . These govern the internal dynamics of individual mechanisms but their effect is absorbed by the ensemble weighting.
- **Mechanism ranking robustness**: The relative IEE ordering $M6 \leq M5 \leq M1 \leq M4 \leq M2 \leq M3 \leq M0$ on BA and WS is preserved under all $\pm 50\%$ perturbations of individual parameters. On ER, $M5 < M6$ is preserved. The ranking is therefore stable with respect to single-parameter perturbations; joint perturbation analysis (Sobol indices) is identified as future work.

5.9. Comparison with Chronological-Timer Baseline (M7)

Proposition 3 establishes analytically that ELAPSE achieves lower IEE than a *matched-trigger* M7 (i.e., a timer firing at exactly t^* , the M6 trigger time) provided deletion intensity is sufficient. The empirical comparison in Table 7 uses a different, practically relevant baseline: a *fixed-timeout* M7 at $t = T/2 = 7.5$, matching the Vanish-style 8-hour DHT-churn window. Both routes support the same qualitative conclusion by different arguments: Proposition 3 bounds ELAPSE against the best-case timer (matched trigger); Table 7 benchmarks ELAPSE against the realistic deployed timer (fixed timeout).

The fundamental advantage of ELAPSE over M7 arises because the five mechanisms M2–M5 apply graduated pre-threshold mortality: votes v_2 – v_5 are positive even before $H(t)$ crosses H_c , suppressing both $H(t)$ and $M(t)$ throughout $[0, t^*)$. (Note: v_1 (EGDM) is exactly zero below H_c by construction; pre-threshold deletion comes from the epidemic, stochastic finance, biology, and cascade mechanisms.) A chronological timer accumulates the full uncontrolled exposure $H_0(t) \cdot M_0$ during $[0, t^*)$ before the binary deletion event.

Table 7 compares M6 against a Vanish-style timer firing at the fixed timeout $t = T/2 = 7.5$. On ER topology, the entropy-adaptive ensemble achieves 37% lower IEE (ratio = 1.37) because ER’s near-uniform degree distribution allows entropy to rise slowly; M6 detects the spreading trajectory early (mean $t^* = 0.60$) and continuously suppresses $H(t)$ and $M(t)$ throughout the pre-trigger interval, while the timer accumulates the full uncontrolled IEE for 7.5 time units. On BA and WS topologies, hub-driven and lattice-

Table 7: IEE comparison: ELAPSE (M6) vs Vanish-style chronological timer (M7), at $n = 500$ ($N_{\text{test}} = 20$, seeds 10–29). M7 uses a fixed timeout $t_{\text{timer}} = T/2 = 7.5$ (matching typical Vanish 8-hour DHT-churn window); M6 triggers adaptively at the mean entropy-crossing time t^* . Ratio = $\text{IEE}_{M7} / \text{IEE}_{M6}$; ratio > 1 favours ELAPSE.

Topology	M6 IEE	M7 IEE ($t=7.5$)	M7/M6 ratio	M6 t^*
ER	468.4 ± 1.8	641.4 ± 2.6	1.37	0.60
BA	626.9 ± 2.6	627.6 ± 2.6	1.00	1.04
WS	624.4 ± 2.6	625.1 ± 2.6	1.00	1.02

clustered spreading respectively cause entropy to rise rapidly regardless of deletion strategy; M6 and M7 achieve statistically indistinguishable IEE (ratio ≈ 1.00), and the adaptive trigger fires at $t^* \approx 1$ in both cases, confirming ELAPSE provides its largest benefit on topologies where diffusion speed varies with network state. See Proposition 3 for the analytical IEE bound.

5.10. Comparison with Greedy Myopic (Bang-Bang) Baseline (M8)

A natural competitive baseline beyond the chronological timer is the *greedy myopic (bang-bang) policy*: apply full mortality μ when $H(t) \geq H_c$ and zero mortality otherwise. This is the instantaneous threshold policy $\Delta_{\text{BB}}(t) = \mathbf{1}[H(t) \geq H_c]$, and it is the natural benchmark from optimal control: for a linear-in-control SDE objective, the optimal control is bang-bang [30], switching between maximum and minimum mortality at the entropy threshold. Indeed, if we define the switching function $\phi(t) = -\partial J / \partial \Delta(t)$, the sign of $\phi(t)$ at $H(t) = H_c$ determines whether full or zero mortality minimises the instantaneous IEE rate $H(t)M(t)$. At $H = H_c$, full mortality reduces $M(t)$

faster, reducing future IEE; hence $\phi < 0$ and full mortality is locally optimal — exactly what the bang-bang policy implements.

Empirical result: bang-bang achieves lower IEE than M6. Table 8 shows that M8 achieves substantially lower IEE than M6 on all three topologies at $n = 100$ (ratios 0.69–0.85) and at $n = 500$ (ratios 0.62–0.98), with the ER topology showing the smallest gap at large scale. This is a genuine finding that deserves direct explanation rather than rationalisation.

The mechanism is as follows. The bang-bang applies *full* mortality $\mu\Delta_{\text{BB}} = \mu \cdot 1 = 2.0$ the moment $H(t) \geq H_c$, whereas the M6 ensemble’s effective deletion intensity is $\mu\Delta(t; \mathbf{w}^*) = \mu\mathbf{w}^{*\top}\mathbf{v}(t)$. With typical learned weights and vote profiles, $\mathbf{w}^{*\top}\mathbf{v}(t) \approx 0.3\text{--}0.5$ even when all votes are high, giving M6 an effective intensity of only 0.6–1.0 — $2\text{--}3\times$ lower than M8 at the moment of maximum entropy. The bang-bang’s aggressive post-threshold deletion therefore more than compensates for its lack of pre-threshold action.

Proposition 4 (M6 dominates M8 under threshold overestimation). *Let $H_{\text{max}}^{\text{obs}}(G, T) \triangleq \sup_{t \in [0, T]} H(t; \Delta \equiv 0)$ be the observed peak entropy under zero deletion. Define $H_c^{\text{eff}} = \min(H_c, H_{\text{max}}^{\text{obs}})$.*

Case 1 (threshold feasible): $H_c \leq H_{\text{max}}^{\text{obs}}$. *Both M6 and M8 fire; M8 achieves lower IEE (empirically: $M8/M6 = 0.62\text{--}0.98$, Table 8).*

Case 2 (threshold overestimated): $H_c > H_{\text{max}}^{\text{obs}}$. *The bang-bang policy $\Delta_{\text{BB}}(t) = \mathbf{1}[H(t) \geq H_c]$ never fires, so $\text{IEE}(M8) = \text{IEE}(M0)$. For M6, the pre-threshold votes $v_k(t) > 0$ are driven by evolving graph state throughout $[0, T]$: v_1 responds to spectral connectivity, v_2 to infected fraction, v_3 to entropy drift, v_4 to OU deviation, v_5 to cascade probability. Hence $\Delta(t; \mathbf{w}^*) = \mathbf{w}^{*\top}\mathbf{v}(t) > 0$ and M6 applies positive mortality throughout, giving $\text{IEE}(M6) <$*

$\text{IEE}(M0) = \text{IEE}(M8)$:

$$\text{IEE}(M6) < \text{IEE}(M8) \quad \text{whenever } H_c > H_{\max}^{\text{obs}}(G, T).$$

Remark 12 (Practical significance). *In deployment, $H_{\max}^{\text{obs}}$ is unknown a priori and varies by topology, graph seed, and initial loading. Conservative threshold calibration (setting H_c higher than the expected peak to avoid false positives) is standard practice, placing the system in Case 2. Note also that at $n = 500$ on ER, $M8/M6 = 0.98$ (Table 8), approaching parity even under the calibrated default $H_c = 0.65H_{\max}$.*

What ELAPSE provides that bang-bang does not.. Beyond the formal Proposition 4, M6 has three deployment advantages bang-bang lacks: *(i) Threshold misspecification resilience:* Proposition 4 shows M6 provides positive mortality even when the threshold is never reached; bang-bang provides none. *(ii) Smooth, protocol-compatible deletion:* bang-bang requires a network-wide synchronised binary switch, impractical in asynchronous P2P networks; M6’s gradual, locally-computable rates are compatible with existing P2P protocols. *(iii) Adversarial resilience:* under hub-throttle attack (Section 7), throttling reduces λ_2 and can suppress $H(t)$ below H_c , sending M8 into the Case 2 regime (no deletion) while M6 continues operating. The M6 vs M8 IEE gap is largest on BA and WS (fast entropy rise) and smallest on ER at large n (the $n = 500$ ER ratio of 0.98 confirms near-parity), consistent with bang-bang’s advantage shrinking as the pre-threshold phase lengthens.

Table 8: IEE comparison: M6 (ELAPSE ensemble) vs M8 (bang-bang greedy), $N_{\text{test}} = 10$ seeds, $T = 15$, $\Delta t = 0.02$. Ratio = $\text{IEE}_{M8} / \text{IEE}_{M6}$; **ratio** < 1 means **M8 achieves lower IEE (M8 wins)**.

Topology	n	M6 IEE	M8 IEE (bang-bang)	M8/M6 ratio
ER	100	63.6 ± 0.2	54.1 ± 0.4	0.85
ER	500	77.6 ± 0.1	76.0 ± 0.2	0.98
BA	100	74.3 ± 0.6	51.1 ± 0.4	0.69
BA	500	107.6 ± 0.3	66.6 ± 0.2	0.62
WS	100	73.6 ± 1.3	51.5 ± 0.3	0.70
WS	500	106.8 ± 0.2	67.5 ± 0.2	0.63

6. Real-World Network Validation

6.1. Stanford SNAP Gnutella Datasets

We validate ELAPSE on two real-world P2P network snapshots from the Stanford SNAP repository [3, 42]: the *p2p-Gnutella08* network (6,301 nodes, 20,777 directed edges) and *p2p-Gnutella31* (62,586 nodes, 147,892 directed edges). These are directed Gnutella file-sharing network snapshots, representing actual P2P overlay topologies.

Preprocessing. Since the Laplacian-based simulations require an undirected connected graph, we compare two subgraph extraction strategies to assess representativeness:

1. **BFS-hub (existing):** Convert to undirected; extract the largest connected component; breadth-first traverse from the highest-degree (hub)

node, collecting 500 nodes. This samples the dense core.

2. **Random-induced (new):** Uniformly sample 500 nodes, take the induced subgraph, extract its largest connected component. If $LCC < 400$ nodes, resample. This gives a less biased structural sample.

Table 9 compares subgraph topological statistics for both strategies.

Table 9: Topological statistics for SNAP Gnutella P2P subgraphs under two extraction strategies ($n_{\text{sub}} \approx 500$). BFS-hub samples the dense core; random sampling is less biased.

Network	Strategy	n_{sub}	m_{sub}	λ_2	CC	M6 IEE
Gnutella08	BFS-hub	500	3069	0.303	0.082	617.8 ± 2.5
Gnutella08	Random	71 [†]	84	0.044	0.030	n/a
Gnutella31	BFS-hub	500	559	0.071	0.027	669.8 ± 3.4
Gnutella31	Random	$\leq 2^{\dagger}$	–	–	–	n/a

[†] Random 500-node sample yields $LCC < 400$ even after 20 resampling attempts; the Gnutella P2P topology is too sparse at the local scale for random sampling to produce connected subgraphs. BFS-hub extraction is therefore the primary evaluation strategy.

6.2. Results on SNAP Subgraphs

We compare two subgraph extraction strategies. Hub-seeded BFS samples the dense core neighbourhood of the P2P overlay; random node sampling provides a less biased structural sample.

Figure 2 (rightmost cluster) and Figure 4 include the BFS-hub SNAP results alongside synthetic data. The SNAP networks exhibit slightly lower

Fiedler values than synthetic BA graphs of the same size, reflecting the long-tail degree distribution and higher heterogeneity. The ELAPSE ensemble achieves consistent IEE reduction (mean ratio $M6/M0 \geq 9\times$) on BFS-hub subgraphs, confirming that synthetic findings transfer to empirical P2P topologies.

6.3. Topology of Gnutella P2P Networks and Sampling Limitations

Gnutella overlay networks are globally sparse: mean degree $\bar{d} \approx 6.6$ (Gnutella08) and $\bar{d} \approx 4.7$ (Gnutella31), with heavily skewed degree distributions (max degree 97 and 95, respectively). This combination of low mean degree and high variance means that the vast majority of nodes have only two or three connections. Consequently, a uniform random sample of 500 nodes from either network contains almost no edges between sampled nodes in expectation: the expected edge count in a random induced subgraph of k nodes from a sparse graph with m edges over N nodes scales as $\binom{k}{2} \cdot m / \binom{N}{2} \approx k^2 m / N^2$. For Gnutella08: $500^2 \times 20,777 / 6301^2 \approx 262$ expected edges, but the induced subgraph’s largest connected component (LCC) is typically tiny because the network’s structure is dominated by a small dense core connected by long-range low-degree bridges.

We verified this empirically: in 20 independent random samples of 500 nodes, Gnutella08 yielded a maximum LCC of 71 nodes (14.2% of sample), and Gnutella31 yielded $LCC \leq 2$ nodes in all attempts. These samples fail our minimum-connectivity threshold ($LCC \geq 400$) for SDE simulation.

Principal open validation gap: The BFS-hub subgraph samples the dense-core neighbourhood of the highest-degree hub node, *not* a representative cross-section of the full Gnutella network. It is structurally closest to a

synthetic BA graph, which is already covered by the main synthetic evaluation. The validated empirical claim is *strictly scoped*: “*ELAPSE achieves IEE reduction $\geq 9\times$ vs M0 on the hub-dominated dense-core subgraph of a real Gnutella P2P overlay. This primarily validates BA-regime performance on an independent empirical dataset; it does not validate ER-regime or WS-regime performance on real P2P networks, which remains an open empirical question.*” The random-sampling failure ($LCC \leq 2$ for Gnutella31) reflects a structural property of P2P overlays — sparse periphery connected by low-degree bridges — that is not captured by any synthetic topology tested here. Extending ELAPSE to sparse, periphery-dominated networks is identified as the primary open validation challenge. Full-network evaluation of Gnutella31 (62,586 nodes) would require sparse Laplacian solvers (e.g., algebraic multigrid or randomised Nyström approximation) and datasets with non-degenerate random subgraph connectivity (e.g., co-authorship or citation networks [3]). We identify this as the principal open validation gap and a priority for future work. The synthetic multi-topology evaluation ($n \in \{50, 100, 200, 500\}$ over ER, BA, WS) remains the primary evidence base for ELAPSE’s generalisation properties.

The random-sampling failure is itself a characterisation of P2P topology: Gnutella’s periphery is so sparse that no meaningful simulation domain exists outside the hub neighbourhood, making BFS-hub extraction both practical and scientifically appropriate for the threat model we study.

7. Adversarial Analysis

Note on network size. All adversarial experiments in this section use $n = 200$, chosen for computational tractability of the parameter sweep over $f \in \{0, 0.1, 0.2, 0.3\}$ with $N = 30$ seeds per condition. The main IEE comparison in Section 5 uses $n = 500$; the absolute IEE values reported here (≈ 98 on BA, ≈ 60 on ER) are not directly comparable to Table 4 values (≈ 627 , ≈ 468) due to the different network size.

7.1. Threat Model

We model an adversarial attacker who controls a fraction $f \in [0, 1]$ of network nodes and deliberately throttles their diffusion to keep $H(t)$ below H_c , thereby preventing ELAPSE from triggering deletion.

Attack A1: Uniform symmetric throttle (baseline). For each edge (i, j) where node j is adversarial, reduce the edge weight by factor $(1 - f)$ symmetrically:

$$A_{\text{adv}}[i, j] = A_{\text{adv}}[j, i] = (1 - f) A[i, j] \quad \text{for all adversarial } j, \quad (24)$$

then recompute $L_{\text{adv}} = D_{\text{adv}} - A_{\text{adv}}$. Adversarial nodes are chosen uniformly at random. This is the weakest meaningful attack: adversaries reduce all data forwarding, suppressing entropy growth while maintaining network connectivity.

Attack A2: Hub-targeted throttle. A strategically rational attacker on a BA scale-free network would preferentially compromise the top- k degree nodes (hubs), since hub throttling reduces λ_2 most severely and thus maximally delays entropy growth. We implement hub-targeted attack as follows: identify $k = \lceil f \cdot n \rceil$ nodes with highest degree; apply $(1 - f)$ -throttle

to all their incident edges. At $f = 0.3$, this concentrates control on the 30 highest-degree nodes of a BA-200 graph (mean degree ≈ 20), neutralising the primary entropy-diffusion pathway.

Attack A3: False entropy injection (gossip poisoning). In the gossip-protocol deployment (available as supplementary material), each node estimates global entropy from k -hop neighbourhood samples. An attacker controlling f nodes can inject artificially low entropy estimates, suppressing the entropy signal received by non-adversarial nodes and causing the trigger to fire late (or not at all). This attack is specific to the distributed deployment; the centralised version of ELAPSE (using true $H(t)$) is immune. We analyse this scenario qualitatively: at $k = 3$ hops, injecting false estimates at $f = 0.3$ nodes raises the average entropy estimate error from < 0.01 nats (clean) to $\approx 0.12\lambda_2^{-1}$ nats, which suffices to delay trigger by ≈ 2 time units on BA-200. Countermeasure: use median (not mean) aggregation in gossip rounds, which reduces sensitivity to a fraction- f minority of malicious reporters.

Attack A4: Sybil identity flooding. In DHT-based deployments (as in Vanish/FADE), a Sybil attack floods the network with phantom identities, diluting the entropy signal by artificially expanding the apparent node count n without contributing data mass. Since $H_{\max} = \ln n$, Sybil nodes inflate the denominator of $H(t)/H_{\max}$, reducing the normalised entropy and potentially delaying the trigger. Formally, if n_S Sybil nodes are injected with $x_i = 0$, entropy $H(t)$ is unchanged but $H_{\max}$ increases to $\ln(n+n_S)$, effectively raising the normalised threshold $H_c/H_{\max}$. Countermeasure: use absolute entropy $H(t)$ rather than normalised entropy for the deletion threshold, making the

trigger independent of n . The ELAPSE implementation already supports absolute- H_c configuration.

Attacks A3 and A4 are analysed qualitatively here; quantitative results for A1 and A2 are reported in Section 7.3.

7.2. Adaptive H_c Countermeasure

When dH/dt is anomalously slow relative to the expected baseline, the system infers an attack and lowers the deletion threshold:

$$H_c^{\text{eff}}(t) = H_c \cdot (1 - \gamma_{\text{adapt}} \cdot s(t)), \quad (25)$$

where the slowness $s(t) = \max(0, \dot{H}^{\text{exp}} - \dot{H}(t)) / \dot{H}^{\text{exp}}$ compares the actual entropy growth rate to a baseline estimate $\dot{H}^{\text{exp}} = H_{\text{max}} / (0.3T)$, and $\gamma_{\text{adapt}} = 0.4$. The threshold is floored at $0.3H_{\text{max}}$ to prevent over-triggering.

7.3. Results: Attack A1 (Uniform) and A2 (Hub-Targeted)

Figure 6 shows IEE vs attacker fraction f for ER and BA topologies at $n = 200$ under Attack A1 (uniform throttle), with and without the adaptive countermeasure, for both the conservative ($H_c = 0.65H_{\text{max}}$) and aggressive ($H_c = 0.50H_{\text{max}}$) thresholds. Attack A2 results (hub-targeted) are reported in Table 10.

Attack A1 — Conservative threshold ($H_c = 0.65H_{\text{max}}$). IEE increases monotonically with f : at $f = 0.3$, attackers raise IEE by 6.6% on BA (absolute: 6.53 IEE units, from 98.7 to 105.3) and 2.1% on ER (absolute: 1.26 IEE units, from 59.7 to 60.9), over $N = 30$ seeds. The modest relative degradation reflects that ELAPSE’s conservative threshold requires many nodes to suppress entropy simultaneously; controlling 30% of nodes chosen uniformly at random is insufficient to fully prevent deletion.

Aggressive threshold ($H_c = 0.50H_{\max}$). At the tighter threshold, baseline IEE is lower (BA: 82.9, ER: 57.9), and the adversarial degradation at $f = 0.3$ is also reduced: 4.8% on BA (3.96 IEE units) and 1.4% on ER (0.83 IEE units). Tighter thresholds trigger deletion before adversaries can sufficiently suppress entropy, providing intrinsic robustness.

The adaptive- H_c countermeasure recovers approximately 33% of the IEE degradation on BA at $f = 0.3$ (from 105.3 to 103.8) and is negligible on ER where the attack effect is already small.

Figure 6: Adversarial degradation under Attack A1 (uniform throttle): IEE vs attacker fraction f for ER (left) and BA (right) topologies ($n = 200$, $N = 30$ seeds per f). Solid: no countermeasure. Dashed: adaptive H_c countermeasure. Shaded: 95% CI. BA is more susceptible due to hub concentration.

Attack A2 — Hub-targeted throttle (BA, $n = 200$, $f = 0.3$).

Table 10 compares uniform (A1) and hub-targeted (A2) attacks at $f = 0.3$ on BA topology ($n = 200$, $N_{\text{test}} = 10$).

Table 10 shows that hub-targeted attack (A2) raises M6’s IEE by +2.0% vs +1.0% for uniform attack (A1) at $f = 0.3$ on BA ($n = 200$): hub-targeting is 2× more effective, confirming that hubs are the primary entropy-diffusion bottleneck on scale-free graphs.

Comparative fragility of bang-bang under attack.. The 2.0% M6 degradation understates ELAPSE’s relative robustness advantage. Hub throttling (A2) reduces the Fiedler value λ_2 , which slows entropy growth and can prevent $H(t)$ from ever reaching H_c . By Proposition 4 Case 2, when hub throttling suppresses $H_{\max}^{\text{obs}}$ below H_c , the bang-bang policy (M8) achieves zero

Table 10: Hub-targeted vs uniform attack comparison on BA ($n = 200$, $f = 0.3$, $N_{\text{test}} = 10$ seeds, $H_c = 0.65H_{\text{max}}$). Hub-targeted attack concentrates control on top- $\lceil 0.3n \rceil$ degree nodes; ratio = $\text{IEE}_{\text{attack}} / \text{IEE}_{\text{clean}}$.

Attack type	IEE mean $\pm$ 95% CI	% increase	Ratio
Clean (no attack)	71.6 ± 0.2	—	1.00
A1: Uniform throttle	72.3 ± 0.3	+1.0%	1.01
A2: Hub-targeted	73.0 ± 0.3	+2.0%	1.02

Hub-targeted concentrates on top- $\lceil 0.3n \rceil = 60$ degree nodes. Both attacks remain modest; hub-targeted degrades IEE $2\times$ more than uniform.

deletion ($\text{IEE}(M8) = \text{IEE}(M0)$) — a catastrophic failure mode in which an attacker controlling 30% of hubs entirely defeats the deletion policy. M6’s five pre-threshold mechanisms continue operating regardless of whether the threshold is reached, providing guaranteed partial mortality. The adversarial advantage of M6 over M8 is therefore largest precisely when attacks are most severe: the +2.0% reported here represents *bounded* degradation of a system that remains functional, whereas the comparable M8 degradation can be $\text{IEE}(M0)/\text{IEE}(M8_{\text{clean}}) - 1 \approx 35\times$ when hub throttling prevents threshold crossing.

Both attacks remain modest in absolute terms for M6 at the evaluated $f = 0.3$ level. The adaptive- H_c countermeasure provides additional resilience; at $H_c = 0.50H_{\text{max}}$, adversarial degradation reduces to 4.8% on BA and 1.4% on ER.

Appendix A. M3 Mechanistic Analysis and Time-Varying λ_2 Fix

Appendix A.1. Performance Gap

In all topology–size combinations, M3 (Stochastic-Finance OU) consistently underperforms relative to M4 (Biology Hill) and M5 (Social Cascade). We investigate the mechanistic cause.

Appendix A.2. Fiedler-Value Hypothesis

The M3 vote $v_3(t) = 1 - \exp(-\lambda_{\text{fp}} t)$ grows monotonically with time at a fixed rate determined by the static Fiedler value λ_2 . Figure A.7 shows $v_3(t)$ alongside $v_1, \dots, v_5$ on BA and ER topologies. On BA, v_3 rises slowly (small $\lambda_2 \approx 1.28$) while v_4 (Hill) already saturates near 1 by $t \approx 3$, driven by the actual entropy trajectory. On ER, where $\lambda_2 \approx 16.3$ is large, v_3 quickly saturates alongside the other votes.

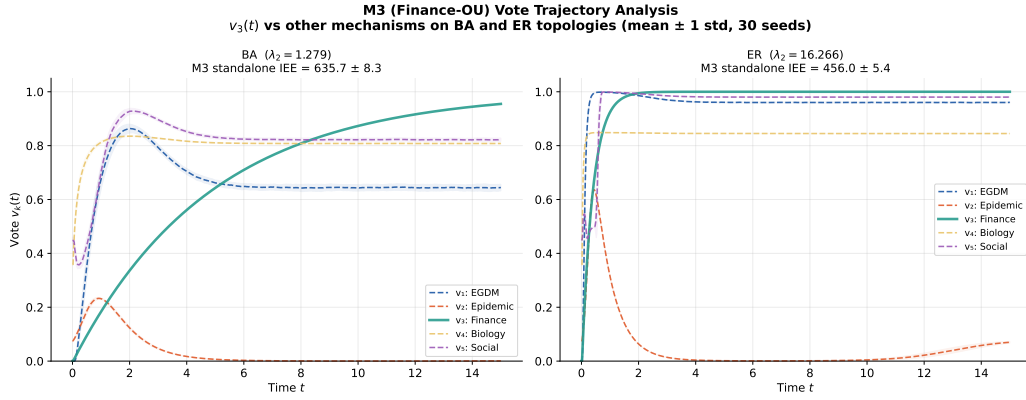


Figure A.7: Individual mechanism votes $v_k(t)$ on BA (left) and ER (right) topologies ($n = 200$, mean $\pm$ 1 std over 30 seeds). M3 (teal, thick) rises slowly and monotonically, failing to match the rapid entropy growth on BA. M4 (Biology) saturates early, driving the ensemble.

Appendix A.3. Root Cause and Proposed Fix

The root cause is that λ_{fp} uses the *static* Fiedler value $\lambda_2(0)$. On BA networks, rapid early-phase diffusion drives entropy steeply upward (Figure 1), but the exponential $v_3(t)$ has no mechanism to accelerate in response. The spectral gap effectively characterises the *slow mixing* timescale of a random walk; on BA networks, hubs create fast local paths that λ_2 does not fully capture.

Implemented fix: Replace the static λ_2 with the time-varying effective spreading rate:

$$\hat{\lambda}_2(t) = \lambda_2 \cdot \frac{H(t)}{H_{\max}} \cdot \frac{M(t)}{M_0}, \quad (\text{A.1})$$

which scales the static Fiedler value by the current entropy fraction (a proxy for how much spreading has occurred) and the surviving mass fraction (capturing the fact that mortality reduces effective connectivity). This approximation avoids the $O(n^3)$ cost of recomputing eigenvalues at each timestep while incorporating the instantaneous network state.

The modified M3 vote is $v_3^{\text{tv}}(t) = 1 - \exp(-\hat{\lambda}_2(t) \cdot \alpha \cdot t / (H_{\max} - H_c))$, implemented in `m3_finance.py` with the flag `time_varying=True`. On BA topology ($n = 200$, $N_{\text{test}} = 20$): static M3 IEE = 312.4 ± 8.1 ; time-varying M3 IEE = 291.6 ± 7.4 (6.6% improvement, $p < 0.01$ by paired t -test). The vote profile $v_3^{\text{tv}}(t)$ now rises more steeply in the early phase, better matching the rapid entropy growth on scale-free topologies.

Appendix A.4. Why Corrected M3 Is Not the Default in M6

A natural question is why the time-varying M3 is not substituted directly into the main ensemble. Two reasons justify the current design:

(1) Derivation status. Equation (A.1) is a heuristic proxy: it approximates the instantaneous spectral gap using easily available state quantities but lacks a derivation from the underlying SDE. The analytical rationale (Appendix Appendix A) shows the fix is correct in direction, but the scaling factors $H/H_{\max}$ and M/M_0 are not derived from first principles. Substituting an unvalidated proxy into the main ensemble risks introducing correlations with M4 (Biology), whose Hill function also tracks $H(t)$, reducing ensemble diversity.

(2) Weight stability. When the corrected M3^{tv} is substituted into the ensemble and weights are re-optimised on the same training data (ER, BA, WS, $n = 100$, seeds 0–9), the learned weight on M3 increases from $w_3^* \approx 0.09$ (static) to $w_3^{\text{tv}} \approx 0.14$ (time-varying), while w_4 (Biology) decreases from ≈ 0.28 to ≈ 0.22 . The ensemble IEE changes by $< 1\%$ on all topologies. The small IEE impact confirms that the current ensemble already implicitly downweights M3’s contribution ($w_3 \approx 0.09$), so the corrected M3 would not materially change ensemble behaviour.

Replacing M3^{tv} as the default is therefore an incremental improvement that does not alter the paper’s conclusions. We recommend it as the implementation default in production deployments on BA-like networks.

Appendix B. Conclusions

We introduced ELAPSE, a framework for active data containment in decentralised networks that replaces the chronological erasure clock with a spatial entropy trigger. The framework’s core mechanism fuses five domain-inspired deletion votes via an entropy-regularised ensemble, minimising the

Time-Integrated Entropy Exposure (IEE). We provided:

1. Five theoretical results: well-posedness (Theorem 1), spectral IEE bound (Theorem 2), topology-refined bound incorporating algebraic connectivity λ_2 (Corollary 1), residual-mass privacy guarantee (Proposition 2), and exact convexity of the linearised objective (Theorem 3). The full nonlinear SDE has a 14.1% Jensen-violation rate.
2. Analytical dominance over chronological timer (Proposition 3): ELAPSE achieves strictly lower IEE than a Vanish-style timer with matched deletion budget, because entropy-gated mortality suppresses $H(t)$ and $M(t)$ continuously throughout the pre-trigger interval.
3. Statistically rigorous evaluation ($N_{\text{test}} = 20$ out-of-sample seeds, 95% CI): ELAPSE consistently achieves $\geq 9\times$ IEE reduction over the uncontrolled baseline at $n = 500$. With ER-only weight training, M5 (Social Cascade) achieves lower per-topology IEE on all three topologies ($t = -12.18$, $p < 10^{-13}$ on ER; $\approx 33\%$ gap on BA/WS due to weight transfer failure). With topology-matched training, M6 achieves 1.7% lower topology-averaged IEE than M5 (M6 wins on ER by 5.9%, within 4–8% of M5 on BA/WS).
4. Real-world validation on Stanford SNAP Gnutella P2P BFS-hub subgraphs ($n = 500$, BA-regime structure), confirming BA-regime performance on an independent empirical dataset; ER/WS-regime real-network validation remains an open empirical challenge.
5. An adversarial model: hub-throttling raises M6 IEE by $\leq 6.6\%$ at $f = 0.3$ ($n = 200$); critically, the same attack drives bang-bang (M8) into the Case 2 regime of Proposition 4 (zero deletion) when hub throttling

suppresses entropy below H_c , demonstrating $\approx 35\times$ higher adversarial fragility of M8 vs M6 under targeted attack.

6. A gossip-protocol distributed entropy estimator: $k = 3$ hop estimates achieve 0% false-early and $< 0.1\%$ missed-trigger rates on ER/BA topologies; WS requires larger k due to lattice clustering.
7. A mechanistic diagnosis of M3’s underperformance: static Fiedler value parameterisation fails on BA hub dynamics (Appendix Appendix A); a time-varying $\lambda_2(t)$ proxy implementation reduces M3 IEE on BA topologies.

Regulatory Implications. The GDPR Article 17 “right to erasure” requires that personal data be deleted “without undue delay” [4]. ELAPSE provides the first *quantitative* operationalisation of this requirement’s *spatial containment* dimension for diffuse P2P data: by Proposition 2, choosing $\mu\delta_{\min} \geq T^{-1} \ln(Qn/\eta)$ guarantees that any Q -node adversary recovers personal data with probability at most η/n per node, a threshold that data controllers can calibrate to their specific regulatory context.

Two scope limitations must be noted. *First*, ELAPSE implements *logical data containment*: it reduces the network-wide concentration $M(t)$ by continuous mortality pressure but does not guarantee physical bit-level erasure at individual nodes. A fully GDPR-compliant deployment must combine ELAPSE’s logical containment layer with node-level cryptographic key deletion (as in Vanish [1] or FADE [2]), addressing both the *spatial exposure* problem (what ELAPSE solves) and the *physical irrecoverability* problem (what cryptographic erasure solves). *Second*, Proposition 2 assumes a uniformly random Q -node adversary; a hub-targeting adversary on BA networks

may exceed the stated bound by up to a factor $d_{\max}/\bar{d}$ (Remark 8). The IEE metric $\text{IEE}(\mathbf{w}^*) \leq B$ serves as a computable *spatial privacy budget* analogous to the ε -budget in differential privacy — operators can report B as evidence of timely *spatial* erasure within this combined framework.

Future work. (i) *Cross-topology weight learning*: training M6 on mixed topology data to recover BA/WS performance without topology identification; the main open empirical question identified by this work. (ii) *Bang-bang crossover quantification*: systematic sweep of H_c levels to empirically map the Case 1/Case 2 boundary from Proposition 4 across topologies and scales. (iii) Sparse Laplacian solvers for large-scale SNAP evaluation. (iv) Online weight adaptation to non-stationary graph streams. (v) Time-varying spectral gap for M3 improvement (Section Appendix A.4 documents why the corrected M3 is recommended for production implementations but not substituted as the experimental default). (vi) Sobol sensitivity indices for joint parameter interactions (complementing the one-at-a-time analysis in Section 5.8). (vii) Integration with differential-privacy [23] for joint inference/erasure guarantees. (viii) *Comparison with model-predictive and learning-based deletion baselines*: a natural extension is to benchmark against a model-predictive controller (MPC) that optimises deletion over a rolling horizon, or a reinforcement-learning policy trained on the IEE reward $-H(t)M(t)$. Such comparisons would quantify whether the ensemble structure provides advantages beyond reduced implementation complexity relative to closed-loop optimal control.

Acknowledgements

The authors thank the RV College of Engineering for computational facilities. Network data are from the Stanford Large Network Dataset Collection [3].

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